

Supply Variabilities in Public Workfares^{*}

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Abstract

This paper presents a model of the labor market where public workfares increase private wages by reducing labor supply. In a dynamic setting, we show that when wages are downwardly rigid, forward-looking employers optimally compress wage increases in response to intertemporal variability in the level of program implementation. The model generates two key predictions: greater variability in program provision results in a larger compression of wage increases, and compression of wage increases is more severe under low inflation. We empirically verify these predictions using data from two large workfares from India.

JEL: I38, J31, J68, O12

Keywords: Labor market, Wage rigidity, Market power, Public workfare, Wage compression, India

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1 Introduction

Public works have long been a standard response of the government in its role as an employer of last resort. Starting with the Poor Employment Act of 1817 in Britain (Blaug, 1963, 1964) and the New Deal Programs during the 1930s in the US (Kesselman, 1978; Bernstein, 1970), large public works have played an increasingly dominant role in providing income support for the poor in many countries. In recent times public workfare programs have been implemented in a number of developing countries in Latin America, Asia, and Africa (Drèze and Sen, 1991; Lipton, 1996; Subbarao, 1997; Keddemman, 1998). The workfare programs in Europe (Lødemel and Trickey, 2001), work-for-dole in Australia (Bessant, 2000), and the government response to Fukushima disaster in Japan (Nagamatsu, 2014) are examples of similar programs that have been implemented for specific groups in developed economies as well.

Apart from their role as safety-net programs, such labor market interventions from the government have been acknowledged to affect key labor market indicators in the private sector as well. For example, public workfare programs can decrease employment in the private sector under perfectly competitive labor markets. On the other hand, if employers have a significant degree of control in the wage determination process, then public workfare programs can potentially increase both employment and wages in the private sector by improving the bargaining power of workers or by increasing worker's reservation wages (Basu et al., 2009).¹ As monopsonistic labor markets are increasingly found to be pervasive (Ashenfelter et al., 2010; Manning, 2011; da Costa and Maestri, 2019), low wage equilibria could exist due to distortions in the labor market and not because of low worker productivity. In such a scenario, workfare programs can not only improve the welfare of program participants but also of non-participants through wage increases in the private sector (Muralidharan et al., 2018).

However, it is important to acknowledge that apart from employer market power, labor markets may also have other types of distortions like downward nominal wage rigidity that may have a strong influence on how public works affect the private labor market. Studies by Kahn (1997); Lebow et al. (1999); Knoppik and Beissinger (2003, 2009); Dickens et al. (2006) and Behr and Pötter (2010) provide evidence of downward nominal wage rigidity in labor markets under different contexts. More recently, Kaur (2019) provides evidence of downward rigidity in nominal wages using wage and employment responses to rainfall shocks in India. Hence, it becomes important to consider the effects of such rigidities on the labor market response to public workfare programs.

Finally, the impact of public workfares depends, to a large extent, on how well they

¹We use the terms public works and workfare programs interchangeably.

are implemented. Successful implementation of public workfares is often hampered by the lack of institutional capacity and budgetary constraints that are most severe for countries most in need of such labor market interventions. The presence of such constraints leads to variability in the level of program implementation.²

This paper provides a theoretical framework to understand how variability in program implementation and the consequent uncertainty can have large negative effects on the gains from public workfare programs. In other words, we show how after controlling for the level of program provision today, uncertainty regarding the level of program provision in the future can dampen or “compress” wage increases. Hence, variability in public workfares can effectively reverse the welfare gains from such policy interventions.

We develop a model of the labor market with a general increasing and concave production function and risk neutral utility functions. These modeling choices lead to results that are general and robust to assuming specific production or utility functions. Our model accommodates three key features of (i) employer market power, (ii) downward nominal wage rigidity, and (iii) temporal variability in the implementation of public workfares. Using a static model of a private labor market, we first show that public workfare leads to a contraction of labor supply for private employment which increases private wages.³ The level of program provision determines the extent of contraction in private labor supply and the consequent increase in private wages. For example, if the public workfare absorbs (say) 5% of the work force then private wages rise by 2%. If the workfare employs a larger proportion of the workforce then private rise further (up to the marginal product of labor).

Next, to accommodate temporal variation in program provision we introduce a two-period model. Keeping the present level of program provision the same, we compare the first-period wage in the static and two-period models. Continuing with the earlier example, if the workfare employs only 1% of the labor force in the second period, the employer would optimally want to reduce wages. This, however, is not possible when wages are downwardly rigid. As a result, forward-looking employers compress wage increases in the first period to be less than the statically optimal wage increase of 2%. Therefore, the compression of wage increases can be attributed to the trade-off between employers’ current and future profits where the level of private wage today is

²We define the level of program implementation or program provision as the actual level of public employment generated by the program. Intuitively, the level of program implementation directly affects the potential gains for i) the intended program beneficiaries, and ii) the non-beneficiaries who stand to gain from the consequent increase in low-skilled private wages.

³The fall in private employment associated with large public workfares is consistent with recent empirical evidence (see for e.g. [Imbert and Papp, 2015](#)).

dependent not only on the current but also future levels of program provision. Hence, if the provision of public workfare deteriorates in the second period with certainty, *ceteris paribus*, the first-period private wage in the two-period model is lower than in the one-period static model.

In reality, however, there is uncertainty regarding the level of program provision in the future. To incorporate this uncertainty, we assume that the level of program implementation or the rate at which government rations demand for work in the program is drawn from a given distribution. Here, greater uncertainty can be associated with an increase in the variance of the distribution that lies between the two extremes of no implementation and complete implementation. In this framework, the optimal wage in the first period is based on optimizing current and expected future profit. We find that greater uncertainty regarding the level of program provision in the future leads to higher compression of wage increases today.

The uncertainty resulting from variability in the program provision could affect wages through other channels as well. Employers could be risk averse and hence reduce investment levels, resulting in lower productivity and lower wages. Uncertainty could also affect choice of goods to manufacture, or affect migration patterns of workers, which could also affect wages. Here, we present a simple model that abstracts away from investment, risk aversion, and migration to show the simplest mechanism through which variability in the provision of public workfares can compress potential wage increases. Adding more complexity to the model may enrich it and demonstrate multiple channels that, under additional assumptions, may give a similar result. Therefore, our model is more general and robust as it relies on fewer assumptions.

Usually, increases in private wages due to large public workfares are expected to negatively affect private employers' profits. In other words, while a decrease in private wages due to a lower level of program implementation may reduce workers' utility, it increases employers' profit. Counterintuitively, we show that a reduction of wages due to increased variability in program provision reduces both employer profit and worker utility. The compression of wage increases reduces employer profit as the marginal gain due to lower private wages is more than offset by the loss in marginal product of labor when wages are not competitively determined.⁴ An important consequence of this finding is that by simply reducing supply variabilities, without necessarily increasing the overall fiscal burden on the state, the government can unambiguously improve overall welfare that results from the accompanying decompression of wage increases.

Our model gives two key empirically verifiable predictions. First, our model pre-

⁴We assume risk neutral preferences for both employers and workers. Incorporating risk-averse preferences only exacerbates the loss in utility associated with variability in program provision.

dicts that the increase in wages due to a public workfare is lower when variation in program provision is high. Second, the compression of wage increases is more severe when inflation is low. This is because the downward nominal wage rigidity constraint binds less often when inflation is high. We use data from two large workfare programs in India to empirically verify these two key predictions. We find that the increase in real private wages due to the workfare is lower as the variability in program provision increases.⁵ In support of the second prediction, we find that the compression of wage increases is significantly higher for low inflation districts relative to high inflation districts.

Our results contribute to the literature in two important ways. First, we supplement the literature on the effects of downward rigidity in wages (Elsby, 2009; Stüber and Beissinger, 2010; Falvey and Kreickemeier, 2009). These studies model wage rigidity through a loss in worker productivity following a downward revision in nominal wages. In these models, firms compress wage increases today in anticipation of future wage cuts that may prove costly to implement. We share the same insight as these studies except our objective is to associate compression of wage increases with the need to cut wages in the future due to a probable decline in program provision; downward nominal wage rigidity is assumed in our model. A similar framework can be used to assess other policy interventions that affect labor market outcomes under downward nominal wage rigidity.

Second, we contribute to the recent theoretical literature on public works by explicitly incorporating temporal variability in the implementation of the public workfares. While recent studies like Basu et al. (2009), Imbert and Papp (2015), and Sukhtankar (2016) acknowledge incomplete implementation in public works and the consequent variable effect on key labor market indicators, our dynamic model allows us to explicitly parameterize the effects of temporal variability in program provision.⁶ On the other hand, while studies like Basu (2013) and Bahal (2016) discuss labor market outcomes under public workfares using models with more than one period, there is no role for variability in program provision as both these studies assume complete program implementation.

The next section discusses our model of public workfare and section 3 discusses some extensions of the model including inflation and perfect competition. Section 4 uses data on large public workfares in India as a case study to present some important empirical regularities regarding the programs and rural labor markets in India, justifying

⁵We use standard deviation in program expenditure as a measure of variability in program provision.

⁶Basu et al. (2009), for example, take the choice of location and the ease of access as an implicit rationing device by the government.

the key assumptions of our model. Section 5 tests the key empirical predictions of the model. Section 6 concludes.

2 Model

We construct a model of a local labor market, say at the level of a village. We assume labor to be of measure one. Wage from private employment is w . The cost of working c is distributed uniformly over the interval $[0, C]$. This cost includes the cost of effort and other costs such as the opportunity cost of not migrating elsewhere. The utility of private employment is given by $u = w - c$. We normalize reservation utility as zero. For those who are employed, $u \geq 0$, which implies $c \leq w$. Labor supply is given by the fraction of people who are willing to work at wage w . In the absence of any other employment, the labor supply in the private sector is a function of the private wage: $l_0(w) = w/C$. Below we make our first assumption.

Assumption 1. Employers have market power.

Labor markets that are most in need of public workfares exhibit high rates of under-employment/unemployment and low wage levels, factors that are indicative of market distortions like employer market power. For example, we discuss evidence of imperfect competition in rural labor markets in India in section 4. Hence, our model assumes employers to have market power. However, in section 3.2 we show even in perfectly competitive markets, variability in program provision can result in compression of wage increases, albeit through a different mechanism. In section 5 we also show that our empirical results are robust to variation in the degree of market power.

For ease of algebra, we assume a monopsonistic labor market structure. However, our results do not depend on the degree of market imperfection and are valid under oligopsony as well (see Bahal and Shrivastava, 2016).

We model the production function as increasing and concave in l . Let the production function be represented by $f(l)$, such that $f'(l) > 0$, $f''(l) < 0$. Normalizing the price of output to unity, the profit function is therefore

$$\pi_0(w) = f(l_0(w)) - w l_0(w)$$

The employer chooses the optimum wage to maximize profit. We can use the first

order condition to get this optimum level, which will be the equilibrium wage.

$$w_0^* = \arg \max_w [f(l_0(w)) - w l_0(w)]$$

2.1 Public workfare

We now introduce a public workfare in the model. We assume that the government offers a wage w_g under the program. The wage w_g is set at a socially acceptable level that is higher than the prevailing market wage w_0^* and the government cannot change it. In reality, even if the government fixes a legal wage, the actual wage paid in the program as well as the resulting market wage are likely determined simultaneously. As we are interested in how the market wage responds to government wage, we take the latter as exogenously fixed. For the converse treatment see [Niehaus and Sukhtankar \(2013b\)](#).

As $w_g > w_0^*$, everyone employed at wage w_0^* , as well as some of those not employed would want to work in the public workfare. In this context, wages represent the net benefit of working. Therefore, while wages paid in public works can be comparable to private wages, the substantially lower cost of working in public workfares in terms of worker qualifications and access to work makes them more attractive than comparable manual labor work in the private sector.⁷

Let e_0 be the expenditure required to employ everyone who wants a job under the workfare. We assume that the actual budget e is lower than e_0 and hence not all demand for work under the workfare can be satisfied. Workers who do not get employed in the workfare despite wanting to work in it are said to be rationed out and the rate of rationing is defined as

$$r = 1 - \frac{e}{e_0}$$

We assume that the government randomly selects people who get employed in the workfare. This means that for workers wanting to work in the workfare, the cost of working c is not correlated with the probability of getting work in the workfare.⁸

The new labor supply for private employment consists of two parts. When private

⁷Having w_g lower than the prevailing market wage in our model would render the public workfare completely ineffective as it will be unable to attract any workers. [Niehaus and Sukhtankar \(2013b\)](#) find that even when the legal government wage was set lower than the prevailing market wage, government officials increased the actual wage paid so that they could get people to work and hence siphon off money through corruption. Here we use wage w_g to represent the net benefit of working in public workfare, allowing us to simplify the model while reflecting the demand for public work that would exist in reality.

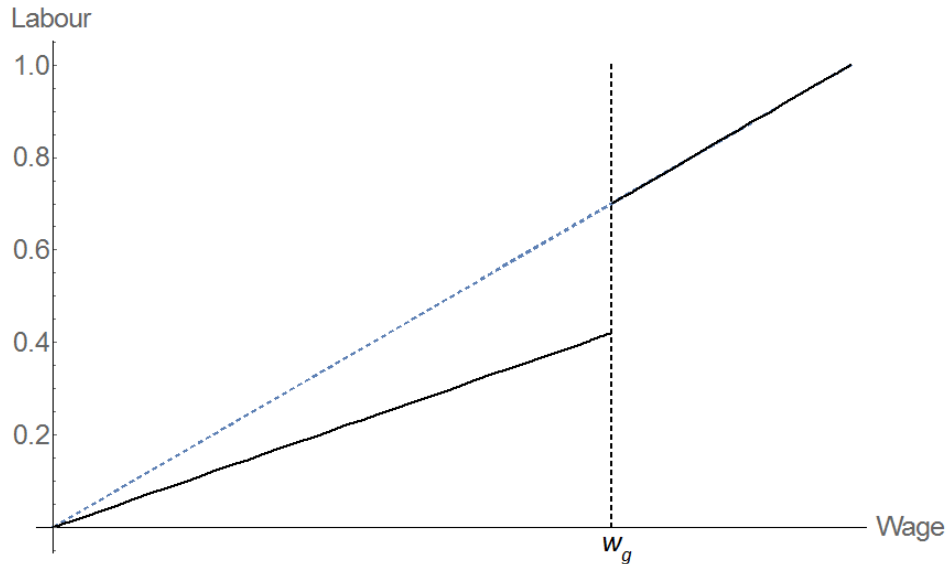
⁸Relaxing the assumption that selection into the workfare is independent of the cost of working does not change our results (see [Bahal and Shrivastava, 2016](#)).

wage w is less than the workfare wage w_g , all workers employed in the private sector (in the absence of workfare) now want to work in the workfare. However, since a fraction $0 < r < 1$ of such workers are rationed out of the workfare, these workers form the new labor supply for the private sector. For $w > w_g$, the situation is exactly the same as with no workfare since all workers would want to work in the higher paying private employment. Therefore, the new labor supply $l_1(r, w)$ is given by

$$l_1(r, w) = \begin{cases} \frac{wr}{C} & \text{if } w \leq w_g \\ \frac{w}{C} & \text{if } w > w_g \end{cases}$$

where figure 1 shows the discontinuous change in private labor supply at $w = w_g$.

Figure 1: Labor supply for private employment with a public workfare with wage w_g



Note: The upward sloping dashed line represents the labor supply when there is no workfare. The solid line is the labor supply for private employment when there is a workfare that offers wage w_g and rate of rationing $r = \bar{r}$. The labor supply only includes the workers being rationed out of the workfare as long as the private wage is below the workfare wage. When the private wage exceeds the workfare wage, the labor supply reverts to the original case as workers no longer prefer working in the workfare.

Assuming $w \leq w_g$, we get the equilibrium private sector wage using the first order condition of the employer's optimization problem.

The profit function is

$$\begin{aligned} \pi_1(r, w) &= f(l_1(r, w)) - w l_1(r, w) \\ &= f(r l_0(w)) - w r l_0(w) \end{aligned}$$

We find the derivative of $\pi_1(w)$ with respect to w

$$\begin{aligned}\pi_1^w(r, w) &= f'(rl_0(w))rl'_0(w) - rl_0(w) - wrl'_0(w) \\ &= r(f'(rl_0(w))l'_0(w) - l_0(w) - w l'_0(w)) \\ &> r(f'(l_0(w))l'_0(w) - l_0(w) - w l'_0(w)) \quad (\because r < 1 \text{ \& } f''(.) < 0)\end{aligned}$$

At $w = w_0^*$, the right-hand side is zero ($\because \pi_0^w(w_0^*) = 0$). Therefore, $\pi_1^w(r, w_0^*) > 0$. This shows that the optimum wage obtained from the first order condition is higher than w_0^* . If this optimum wage exceeds w_g , then the employer restricts wage at w_g as the labor supply increases discontinuously at that point, and there is no need to increase wage further. Hence, the optimal wage with workfare is:

$$w_1^*(r) = \min \left\{ \arg \max_w \pi_1(r, w), w_g \right\} > w_0^*$$

This shows that private sector wage increases as the workfare is implemented. We can also see that in the case where $w_1^* < w_g$, $\frac{dw_1^*}{dr} < 0$, which implies $\frac{dw_1^*}{de} > 0$.⁹ Hence, ceteris paribus, as the government expenditure increases, rationing decreases and the wage in the private sector increases. Also, while private employment decreases, overall employment increases as the workfare not only employs workers who left private employment but also employs some workers who were previously unemployed in the private sector.

For simplifying the rest of the analysis, we will assume that the workfare wage w_g is always higher than the optimal private wage w_1^* in the presence of the workfare. Since w_g is fixed and the analysis does not seek to find the impact of changes in the workfare wage on the private wage, relaxing this assumption will not affect the results obtained. Hence, we can now write the optimal private wage in the simplified form.

$$w_1^*(r) = \arg \max_w \pi_1(r, w) > w_0^* \quad (1)$$

2.2 Two-period Model

We now move to a two-period model of the labor market with workfare. As our intention is to model a *variable* public workfare, we allow the rationing rates in the two periods to be different. Let the rationing in the first period, t , be r_t . Let us assume

⁹ $\pi_1^w(r, w_1^*(r)) = 0$. Hence, using implicit function theorem $\frac{dw_1^*(r)}{dr} = -\frac{\pi_1^{wr}}{\pi_1^{ww}} \Big|_{w=w_1^*(r)}$ Since $w = w_1^*(r)$ is a maximum, $\pi_1^{ww} < 0$. Using $f''(.) < 0$ it is straightforward to show that $\pi_1^{wr} < 0$. Q.E.D. It is evident that at $r = 1$, this should converge to the private wage without the workfare.

for now that the employer is naive and sets the wage in the first period at the statically optimum level described in the previous section. If the rationing rate in the second period ($t + 1$) is lower than that in the first period, then labor supply would decrease and the optimum wage would be higher than that in the first period. Conversely, if the rate of rationing increases then the optimum wage would be lower. Here we introduce the downward rigidity constraint on wages.

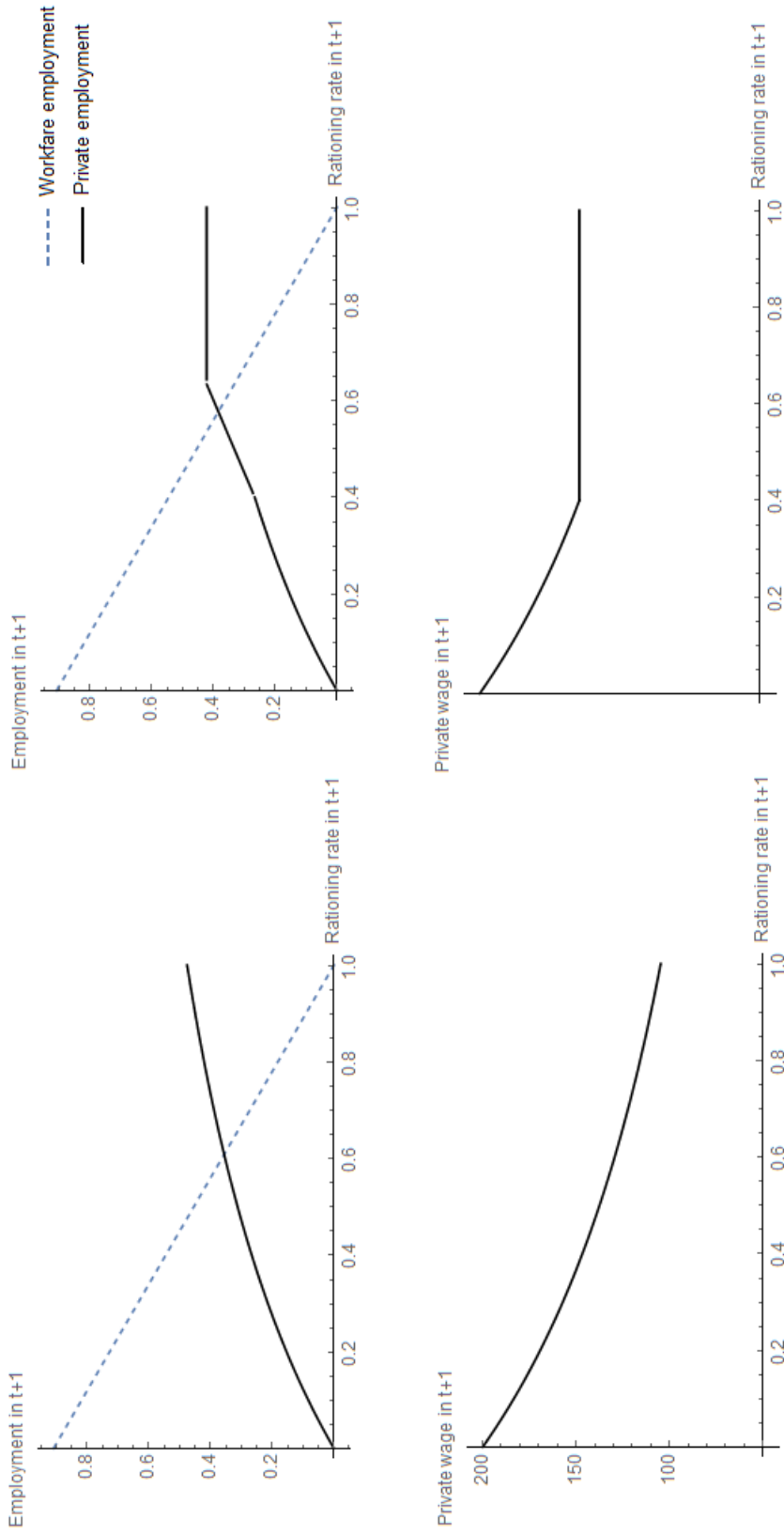
Assumption 2. Private wages are downwardly rigid i.e. wages cannot decrease from period 1 to 2.

There is empirical evidence for this in various labor markets (Kahn, 1997; Lebow et al., 1999; Knoppik and Beissinger, 2003, 2009; Dickens et al., 2006; Behr and Pötter, 2010) as well as theoretical models incorporating wage rigidity (Elsby, 2009; Falvey and Kreickemeier, 2009; Stüber and Beissinger, 2010; Kaur, 2019). We are simplifying the treatment of downward wage rigidity in two ways. First, we are considering perfect rigidity, i.e. wages cannot decrease by construction. A more complex model would introduce a cost of wage reduction – by reducing worker productivity because of perceived unfairness for example. Here we abstract from this by assuming that such costs are large enough to prevent any downward revision of wages. Second, we are abstracting away from price changes and differences between real and nominal wage changes. We will show later that our results continue to hold, albeit more weakly, in the case of price inflation.

Having set the first-period wage at the statically optimum level, the naive employer observes the rationing rate in the second period and tries to maximize profit given the rigidity constraint. Figure 2 shows how the employment and wage in period $t + 1$ change with the rate of rationing r_{t+1} , in the presence and absence of downward wage rigidity.

The graphs on the left show the “no rigidity” case. As r_{t+1} increases, the workfare employment decreases linearly. The private employment increases and private wage decreases as given by the expressions for l_1^* and w_1^* , the optimal employment and wage in the static case. The graphs on the right show the case with downward wage rigidity, with rationing in the first period $r_t = \bar{r}$. Here, the private employment increases and private wage decreases as before till $r_{t+1} = \bar{r}$. At this point, the wage w_{t+1} equals the wage in period t , w_t , and any further increase in r_{t+1} cannot be accompanied with any further reduction in wage to the optimal level. As the rationing rate increases and $r_{t+1} > r_t$, more workers are rationed out of the workfare. Some of these workers whose costs are low enough are employed by the private employer. Hence, private employment increases linearly. As r_{t+1} keeps increasing it is possible that at some point the number

Figure 2: Employment and wages in the second period, with and without downward wage rigidity



Note: The dashed sloping line in both employment graphs is the employment in the public workforce which mechanically decreases from 1 to 0 as rationing increases from 0 to 1. The solid lines on the top and bottom graphs are the employment and wage in the private sector in period $t + 1$. The graphs on the left show the case without downward nominal wage rigidity where private employment, as well as private wage, are continuous functions of r_{t+1} as the employer adjusts both the variables optimally. The graphs on the right show the case with downward nominal wage rigidity with the rationing rate in the first period as $r_t = \bar{r}$, and the wage in the first period being set as optimal for that period only. Both private wage and private employment for $r_t < \bar{r}$ are exactly the same as the case without rigidity. After that, the wage rigidity constraint binds and the employer is unable to lower the wage further. Hence, the private employment increases linearly absorbing a fixed fraction of the workers rationed out of the workforce. After a point, the employer no longer finds it profitable to hire more labor at w_t . Private employment stagnates at this level and any workers who are rationed out because of a further increase in rationing become unemployed.

of workers rationed out of the workfare is higher than the optimal labor requirement of the private employer with the wage fixed at w_t . After this point, private employment no longer increases although the workfare employment keeps decreasing.

Now we consider the case of the forward-looking employer who takes into account the wage rigidity and the variability of rationing rates and hence optimizes over two periods rather than just one. Let us first assume that the employer knows in advance what the rationing rate is going to be in the second period. In the next subsection, we will consider a more realistic case where the employer is uncertain about future rates of rationing. In the present case, the employer maximizes the sum of the profits in the two periods.

$$V(r_t, w_t, r_{t+1}, w_{t+1}) = \pi_1(r_t, w_t) + \pi_1(r_{t+1}, w_{t+1})$$

The profit maximization is done under the rigidity constraint: $w_t \leq w_{t+1}$. The optimal value of w_t depends on r_{t+1} . If $r_{t+1} \leq r_t$, then the wage constraint does not bind, and w_t is given by the expression for static wage w_1^* given in equation 1. If $r_{t+1} > r_t$, then the constraint binds and the second-period wage can no longer be set at the statically optimal level. Since there is a trade-off between profits in the two periods, the optimal w_t in the two-period case will be lower than the corresponding value of w_1^* for the single period case.

Proposition 1. In the presence of downward wage rigidity, the optimal wage in the first period w_t^ is compressed when employment provision under the workfare worsens in the second period.:*

$$w_t^*(r_t, r_{t+1}) < w_1^*(r_t) \quad \text{for } r_{t+1} > r_t$$

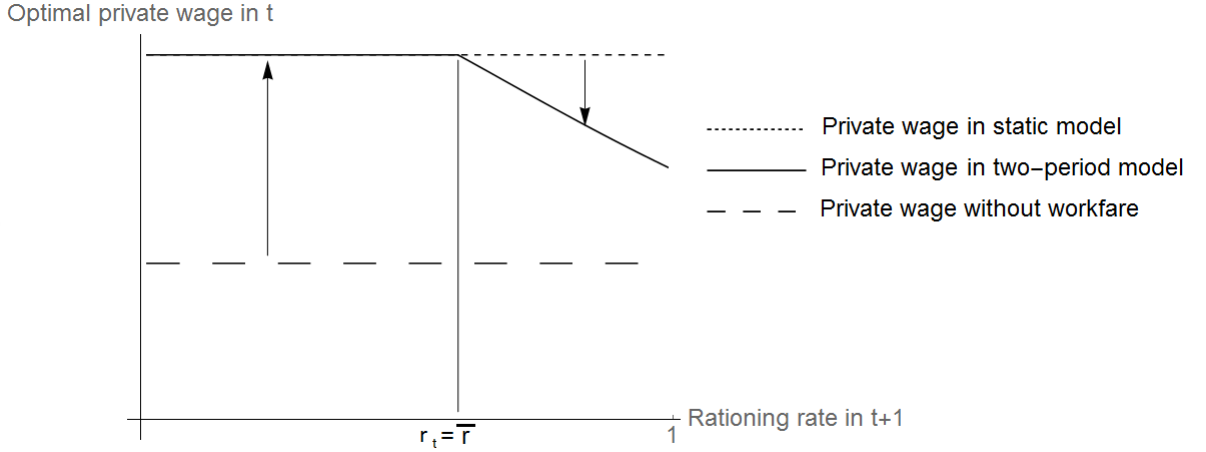
The proof is given in the Appendix. As shown earlier using the implicit function theorem and the concavity of $f(\cdot)$, it is straightforward to see that

$$\frac{\partial w_t^*(r_t, r_{t+1})}{\partial r_{t+1}} < 0$$

Figure 3 plots how the optimal wage in period t , $w_t^*(r_t, r_{t+1})$, changes with r_{t+1} , keeping the rationing rate in the first period constant ($r_t = \bar{r}$). As the figure shows, as long as $r_{t+1} < \bar{r}$, the optimal wage $w_t^*(r_t, r_{t+1})$ is equal to the statically optimal wage $w_1^*(\bar{r})$. However, at $r_{t+1} \geq \bar{r}$, the downward wage rigidity constraint starts to bind as the statically optimal level of the second period wage (not shown in the figure) becomes less

than $w_1^*(\bar{r})$. From this point onwards, the forward-looking employer starts to reduce the first period wage to optimize total profit. Hence, for $r_{t+1} \geq \bar{r}$ an increase in r_{t+1} leads to a compression in the first period optimal wage, represented by a downward sloping line in the figure.

Figure 3: Wage in the three models showing compression of wage increases



Note: The graph shows the increase in private wage due to the workfare in the static case as the gap between the optimal private wage without workfare and optimal private wage with workfare in the static model. The solid line is the optimal private wage in the two-period model, showing how the increase in private wage gets compressed once the rationing rate in the second period r_{t+1} exceeds r_t .

2.3 Uncertainty

Up to now, we have assumed that the employer has perfect information about the rate of rationing in the next period. In reality, however, variability in program implementation would lead to uncertainty associated with forecasting the level of program provision in the future.

Assumption 3. There is uncertainty about future rates of rationing.

Uncertainty in the two-period model implies that there is always a probability that the rate of rationing may be higher in the future. If the employer maximizes the sum of current profit and expected future profit, then the same mechanism as the previous case will lead to compression of wage increases.

Proposition 2. There is a decrease in the optimal wage in the first period, w_t^ , when we move from the case when it is known with certainty that rationing is going to be the same in both periods i.e. $r_{t+1} = r_t$, to a case where there is uncertainty such that*

rationing may go up with some positive probability, i.e. $P[r_{t+1} > r_t] > 0$.

We show the proof in the Appendix but the intuition is evident. As soon as there is a positive probability that the optimal wage in the second period is going to be lower, then the wage rigidity binds and the statically optimal wage in the first period is no longer optimal for both periods. Therefore, the new optimal wage is lower, reflecting the trade-off between profit in the current period and the expected profit in the next period. Proposition 2, therefore, deals with the extensive margin of uncertainty.

Now we look at the intensive margin, i.e. what happens when the degree of uncertainty increases. To do this, we need to define what we mean by an increase in uncertainty. Let the employer believe that the rationing rate is drawn from a distribution. We can think of uncertainty as the variance of this distribution. A distribution with zero variance would imply a fixed rationing rate and would correspond to certainty as discussed earlier. As the variance increases the uncertainty about future rationing increases. For simplicity, let us assume that r is uniformly distributed with support x , given by $U[m - x/2, m + x/2]$, where m is the mean of the distribution. Increasing x increases variance while preserving the mean, hence we can use x as a measure of uncertainty. Our objective is to see how an increase in x changes the optimal wage.

For any given value of w_t , let $\bar{r}_{t+1}(w_t)$ be that value of rationing in the second period where the constraint will just bind if the employer tries to set the statically optimum wage in the second period, i.e. $w_1^*(\bar{r}_{t+1}(w_t)) = w_t$. Hence, in the second period, for a rationing rate less than this value, the employer will set the wage optimally. If the rationing rate exceeds this value, then the constraint will bind and the second period wage will be stuck at the wage set in the first period.

$$w_{t+1} = \begin{cases} w_1^*(r_{t+1}) & \text{for } r_{t+1} \leq \bar{r}_{t+1}(w_t) \\ w_t & \text{for } r_{t+1} > \bar{r}_{t+1}(w_t) \end{cases}$$

We consider the total profit as the sum of current period profit and future expected profit. This will be a function of r_t , m , x and the choice variable w_t .

$$V(m, x, r_t, w_t) = \pi(r_t, w_t) + \frac{1}{x} \left(\int_{m-x/2}^{\bar{r}_{t+1}(w_t)} \pi(r_{t+1}, w_1^*(r_{t+1})) dr_{t+1} + \int_{\bar{r}_{t+1}(w_t)}^{m+x/2} \pi(r_{t+1}, w_t) dr_{t+1} \right)$$

The profit maximizing value of w_t is a function of r_t , m and x . The objective is to understand how this dynamically optimal wage, $w_t^*(r_t, m, x)$, changes with an increase in the variance, x .

Proposition 3. The profit maximizing value of current wage decreases with an increase in the variance in rationing rates, i.e. $\frac{dw_t^}{dx} < 0$, if the profit function $\pi(r, w)$ is such that at $w_t = w_t^*$.*

$$\pi^{w_t}(r_t, w_t) < -\frac{\pi^{w_t}(m + x/2, w_t)}{2} \quad (2)$$

We show the proof in the Appendix. The left-hand side of the inequality in equation 2 represents the marginal profit in the first period that the employer would lose by reducing wage w_t . The right-hand side depicts the worst case scenario in the second period, where the rationing is the maximum possible value, thus creating the maximum gap between the low optimal static wage at this level of rationing, and the actual first period wage w_t , that is stuck because of rigidity. The numerator represents the marginal increase in second period profit if w_t was decreased (a reduction in w_t would bring it closer to the second period statically optimal wage thus increasing profit). The 2 in the denominator comes from the fact that a unit increase in the range x of the distribution increases the worst case rationing by half a unit.

The proposition can thus be interpreted as saying that an increase in variability will compress wages if the reduction in profit in the first period because of reducing the wage is less than half of the consequent increase in profit in the worst case scenario in the second period.

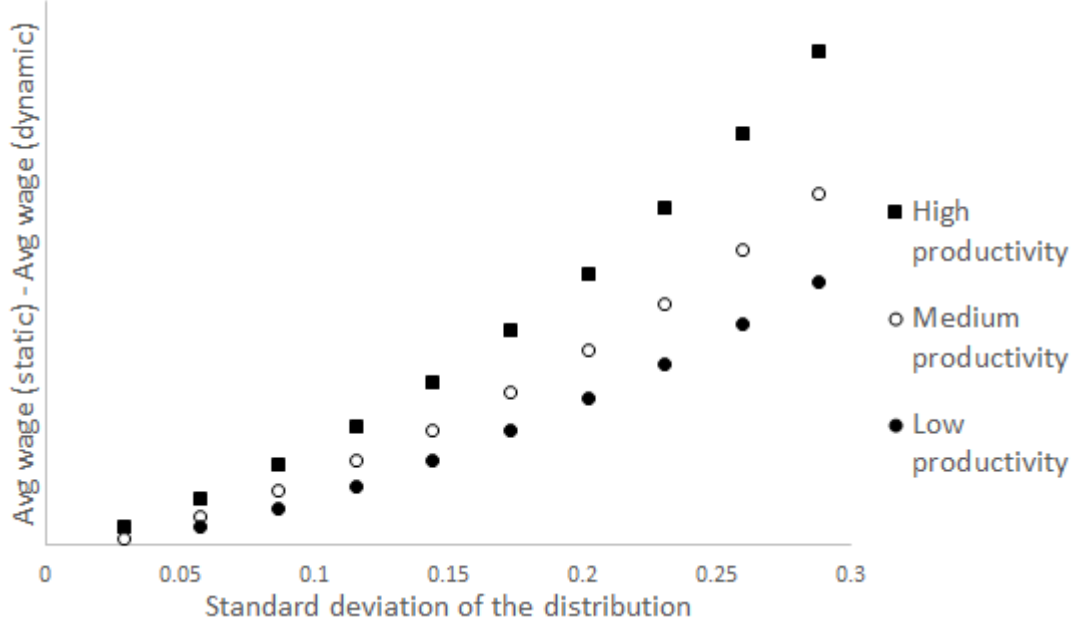
It is evident that if the program is implemented badly in the first period itself, i.e if r_t approaches the maximum value $m + x/2$, the condition is not satisfied. Conversely, there will be a range of values of r_t starting from the lower end of the distribution, for which the condition will hold. We are not particularly interested in whether or not wage increases get compressed at a particular value of r_t . Rather, it is more important to see what the average change in wage is across the entire distribution of r_t .

Determining whether wage compression happens on average becomes analytically intractable. Hence, we conduct simulations where we compute the average optimal wage for different distributions with m fixed at 0.5 but changing values of x . We use a quadratic production function for simplicity.¹⁰ Figure 4 shows the results of the simulations for high, medium and low productivity. We find that the average optimal wage (dynamic) obtained from maximizing the sum of current and expected future profits, is always lower than the average optimal wage (static) obtained by maximizing just the current profit. More importantly, we also find that this difference increases with an

¹⁰ $f(l) = a(l - l^2/2)$, where a is a productivity parameter.

increase in variance signified by an increase in x .

Figure 4: Graph showing increase in wage compression with increase in variance



Note: The horizontal axis shows the standard deviation, which is equal to $x/2\sqrt{3}$, of the uniform distribution with mean $m = 0.5$. The vertical axis is the difference between the statically optimal and dynamically optimal wages averaged over the entire distribution of r_t under the workfare.

Hence, we have shown that increasing uncertainty can lead to increased compression of workfare-led wage increases. While average expenditure in the workfare is directly correlated with the mean rationing rate as higher average expenditure implies lower mean rationing, the variance of rationing is related to the variance of expenditure (keeping the workfare wage constant). This is an important result as it suggests that the government can achieve higher private wages and employment by simply lowering the variability in the provision of public employment while keeping the average expenditure constant.

3 Discussion

3.1 Inflation

In the analysis above we have abstracted away from price changes. The empirical observations regarding downward rigidity are in *nominal* wages. Price inflation changes the wage rigidity condition. Let all the variables above signify real values. Let the price

level p be equal to 1 in the first period, and $p > 1$ in the second period respectively. The rigidity condition would now be $w_t \leq pw_{t+1}$. The only difference this would make in the analysis above is to change the range of values of r_{t+1} for which the constraint would bind. Hence, now the definition of $\bar{r}_{t+1}(w_t, p)$, the threshold value beyond which the constraint will bind will be given by $p w_1^*(\bar{r}_{t+1}(w_t, p)) = w_t$.

Since $\frac{dw_1^*(r)}{dr} < 0$, therefore higher p would lead to higher $\bar{r}_{t+1}(w_t, p)$ for a given w_t . Now consider the total profit function that is maximized by choosing w_t^* .

$$V(m, x, r_t, w_t, p) = \pi(r_t, w_t) + \frac{1}{x} \left(\int_{m-x/2}^{\bar{r}_{t+1}(w_t, p)} \pi(r_{t+1}, w_1^*(r_{t+1})) dr_{t+1} + \int_{\bar{r}_{t+1}(w_t, p)}^{m+x/2} \pi(r_{t+1}, w_t) dr_{t+1} \right)$$

There will be a maximum value of p , say p_{max} , where $\bar{r}_{t+1}(w_t, p) = m + x/2$. In this case the second integral, where the constraint binds, would disappear and the expressions would be

$$V(m, x, r_t, w_t, p) = \pi(r_t, w_t) + \frac{1}{x} \left(\int_{m-x/2}^{m+x/2} \pi(r_{t+1}, w_1^*(r_{t+1})) dr_{t+1} \right)$$

The second period profit now is independent of w_t , hence the optimum value w_t^* is the same that maximizes $\pi(r_t, w_t)$, which is the statically optimum value. In this case, there would be no compression of the wage increase.

The opposite case is when $p = 1$, when we return to the situation described in section 2.3 where the range of the second integral, where the constraint binds, is largest and there is maximum compression of wage increase. Hence, for any value of p between 1 and p_{max} , the compression will keep decreasing as p , i.e. inflation keeps increasing.

This gives us an empirically testable prediction, which says that the wage compression because of increased variability is likely to be lower when inflation is higher. We test and confirm this prediction with Indian data in section 5.

3.2 Perfect competition

We have assumed monopsonistic labor markets in our model above. This is in accordance with the research that finds that agricultural labor markets in developing countries like India are likely to be monopsonistic (see section 4.1 for a more detailed discussion on this). This section shows that the key results coming out of our model hold even if we assume perfect competition in the labor market. Below, we show how compression

of wage increases occurs even with perfectly competitive labor markets, albeit with a different mechanism.

Under perfect competition, the employers demand labor until the marginal product of labor is equal to the market wage.

$$w_{PC} = f'(l_0)$$

Private labor supply is given by a share l_0 of unit measure of workers who choose to work at wage w , where

$$l_0 = w_{PC}/C$$

Hence, the market wage is given by

$$w_{PC}^* = f'(w_{PC}^*/C)$$

Now, if the workfare is implemented with a rationing rate r , only a share r of the workers l_0 willing to work at wage w_{PC} , will actually work in the private sector. Hence, the market wage is given by

$$w_{PC}^* = f'(rw_{PC}^*/C)$$

Since $0 \leq r \leq 1$ and $f''(.) < 0$, as program implementation improves, the market wage increases i.e. $\frac{\partial w_{PC}^*}{\partial r} < 0$. Therefore, similar to the case when the employers have market power, workfare increases private wages even in the presence of perfect competition.

Now, let us allow the rationing rate to be variable. There is no forward looking behavior possible on the part of the employers as they cannot set wages. But if downward wage rigidity exists, it can still lead to distortions since wages cannot adjust downward if workfare implementation deteriorates in the future. As [Elsby \(2009\)](#) notes “... *even in the absence of forward-looking behavior, downward wage rigidity raises the level of wages that firms inherit from the past. As a result, firms do not have to raise wages as often or as much to obtain their desired wage level.*”

To make the above exposition clear, let us imagine two regions with perfectly competitive labor markets that have had the same average levels of program implementation in the past, but with different variance. This would imply that in the region with high variance, the rationing levels are likely to have fallen further below the mean levels, and hence the wage would be stuck at a higher level, compared to the region with low variance. Now let the average level of program implementation improve in both regions in the current period. As wages increase to a new competitive level, the increase would

be lower in the region with high variability.

Hence, in a perfectly competitive labor market, compression of wage increases occurs if the level of wages that employers inherit from the *past* is already high. In monopsonistic markets, it is the uncertainty about *future* program implementation that matters most as employers have wage-setting power and maximize profit intertemporally. To the extent future uncertainty is determined by past variability, the two sets of predictions are identical. In section 5.4.1, we use data on land concentration as a measure of market competition to show that the compression of wage increases due to variability in program expenditure is not significantly different between districts with high and low levels of land concentration.

3.3 Welfare

We now look at how the compression in wage increases impacts welfare. The social welfare function consists of three main components representing the utility of the employer, the utility of the workers, and the cost of the program. Additionally, we would also like to incorporate the social planner's distributional preference over the relative importance of the utilities of the employer and the workers. This can be done in two ways. One is to attach weights directly to the utilities of the employer and the workers. The other way, as used by Basu, Chau, and Kanbur (2009), is to not weight the utilities but have an additional weighted component representing employment, to account for the social planner's concern with increasing employment using a public workfare. The results shown in this section would be valid for either formulation of social welfare but we follow Basu, Chau, and Kanbur (2009) as the implementation of workfare by a government indicates its interest in increasing employment. We define the social welfare function as consisting of the following parts

1. The utility (profit) of the employer (π)
2. The utility of the workers (W)
3. The benefit of increasing employment (ηE)
4. The cost of implementing the program (μB)

Note that as we are using risk neutral utility functions, all four parts can be directly interpreted in money terms.

$$SW = \pi + W - \mu B + \eta E$$

where, $E = (l_p + l_g)$; l_p and l_g are fractions of the labor pool employed in the private sector and the public workfare respectively, $\eta > 0$ signifies the marginal social benefit of employment in addition to the direct benefit through the workers' utility, $B = w_g l_g$ is the money spent on wages paid in the workfare, $\mu > 1$ is a multiplier that incorporates as a proportion of the wage bill, the administrative cost of implementing the program, as well as the marginal cost of public funds to account for distortionary taxation. Let the case when there is no workfare be denoted by the subscript 0. Social welfare is the sum of π_0 , W_0 and ηE_0 . As there is no workfare the cost μB is zero.

Now, let us consider a workfare with a fixed rationing rate. In this case there is no uncertainty and employers optimize statically. We assume that the rationing rate is $\bar{r}$ and that the workfare wage is set at $w_g > w$. The profit π_1 expectedly reduces as the workfare increases wage and decreases employment in the private sector. For workers, since the current private wage as well as the workfare wage are higher than the private wage under the no-workfare case and the total employment is higher as well, it is evident that W_1 and E_1 will increase with the introduction of the workfare. Hence, the introduction of workfare reduces welfare through lower employer profit and increased cost to the government, and increases welfare through increased worker utility and increased employment. Whether this increases or decreases the total welfare depends on the values of the two parameters μ and η , which depict the inefficiency of the government and the social weight attached to achieving higher employment.

We are now interested in comparing a workfare with a fixed rate of rationing, to a workfare with an uncertain level of implementation. With uncertainty, the employer no longer knows the future level of rationing. Instead, the current optimal wage is based on a distribution of probable future rationing rates. For comparison with the previous case, we assume that the distribution has the same mean rationing rate and it represents an increase of variance from zero. We have shown that uncertainty leads to the compression of the wage increase possible under certainty, and also that the compression is higher with increased uncertainty. Since private wage is lower under the uncertainty case, private employment will also be lower, while everything else like the workfare wage and the average rate of rationing remains the same. Hence worker utility W_2 and total employment E_2 will be lower than W_1 and E_1 respectively. The average expenditure remains the same $B_1 = B_2$.¹¹

Now let us consider the employer's profit. Compared to the case with a fixed rationing rate, both private wage and private employment are lower. But employer's profit under uncertainty is unambiguously lower relative to the case without uncertainty (where

¹¹We are assuming that reducing variation does not require any additional expenditure. This may not be true and in reality, there may be a trade-off.

private wage and employment were set at the statically optimal level). This is because the reduction in production due to a reduction in employment is not offset by the reduction in wages. This is expected since wage is always less than the marginal product of labor when employers have market power. The employer is essentially sacrificing profit in this period to counter the expected loss in the second period if program implementation worsens and the wage rigidity binds. In the previous case, this eventuality will never arise as rationing rate is fixed and the employer has no reason to reduce the current period profit. Therefore $\pi_2 < \pi_1$.

Hence, regardless of the values of the parameters μ and η , uncertainty will lead to loss of welfare i.e $SW_2 < SW_1$. Furthermore, we can use the simulation results presented in section 2.3 to extend the analysis done above and state that an increase in uncertainty, represented by an increase in the variance of the distribution of r , reduces welfare as it reduces both private wage and private employment. Hence, the government can improve welfare by simply reducing the variation of program expenditure, without changing average expenditure. This is an important result as it has been found that private employers are generally opposed to the public workfares and improvements in them because this reduces employers' profits (Muralidharan et al., 2018). In this case, however, an improvement in public workfare through a reduction in program variability is beneficial for both employers and workers and is hence likely to have widespread political support.

3.4 Demand driven public workfare

Here we address the concern that uptake of public workfares may not always be supply driven and may be a response to endogenous demand-driven shocks to the private labor market. Suppose the government uses the program to maintain a constant total employment level. Therefore, the program expenditure and hence the rationing rate respond to changes in private employment. The changes in private employment will be driven by changes in private wage which in turn are affected by changes in productivity. While all these variables are determined simultaneously, an equilibrium with higher rationing will be the one where productivity, and hence private wage, is high.

Our objective is to understand if employers will still attenuate their response to the program by compressing the corresponding wage increase in the first period if the variability in program provision is simply a response to productivity shocks. Note that we are not concerned with the employer's response to productivity changes directly (that would happen even without the workfare).

When the variability in program provision is supply driven or exogenous, the em-

ployer compresses wage increase today considering the possibility that the program implementation may deteriorate in future, in which case it will not be possible to re-optimize and reduce the second-period wages because of downward rigidity of wages.

When the level of program implementation changes only to maintain a constant level of total employment, then the only reason for the government to reduce employment in the workfare (increase rationing in the second period) is if the private demand for labor is higher in the second period. This is only possible if the private wage is higher in the second period. But if this is the case, then wage rigidity does not bind and the employer is free to re-optimize (the private wage in the second period w_{t+1} , given the initial level of rationing would have been even higher if rationing was not increased). Hence, there is no reason for the employer to compress the wage increase in response to the workfare in the first period.

This exercise shows that if the public workfare is truly demand-driven, and changes in program expenditure are purely in response to productivity shocks, then there should be no compression of wage increases due to variability in program provision even in the presence of downward wage rigidity.

4 Application to employment programs in India: Descriptive Evidence

In this section, we support the key assumptions made in our model by highlighting some key empirical regularities regarding i) the structure of the labor market, ii) the supply variabilities in implementing large public workfares, and iii) the downward nominal wage rigidity. We use data from two large public workfares in India - Sampoorna Grameen Rozgar Yojana (SGRY) and the National Rural Employment Guarantee Act (NREGA). SGRY was implemented in 2001 and NREGA in a phased manner from 2006 to 2008. Both these schemes were strikingly similar - they were nationwide schemes that provided wage employment through labor-intensive public works and were primarily funded by the central government. Further, both schemes had similar accounting and fund allocation processes.¹² NREGA, therefore, can be understood as an intensification of the pre-existing program SGRY.

First, we characterize the structure of agricultural labor market at the village level in India to support our assumption of employer market power. Next, using data from SGRY and NREGA, we show that such programs are often implemented only partially where the government can ration a substantial proportion of the total demand for work

¹²For a detailed description of the two programs, see [Bahal \(2016\)](#).

under the program. We then discuss temporal variation in the level of program provision to highlight that employment provision is susceptible to deterioration in the future. We further show that the level of program provision today is not very informative of the level of program provision in the future. This underscores the uncertainty associated with partially implemented public workfares. Finally, we present evidence of downward nominal wage rigidity in rural labor markets in India.

4.1 Employer market power

Following the literature on rural agricultural labor markets in India – where public workfares are usually implemented – perfect competition is a strong assumption. In a recent study, [Muralidharan et al. \(2018\)](#) find that private employment did not fall as private wages rose in response to an improved implementation of National Rural Employment Guarantee Act (NREGA) in rural Andhra Pradesh, a result that is incompatible with perfectly competitive labor markets.¹³ There is also qualitative evidence pointing to the existence of market power in rural labor markets ([Ghosh, 2007](#); [Griffin, 1979](#)). [Bardhan \(1984\)](#) notes: “... in about 45 percent of our sample villages (in West Bengal) 7 or fewer employers account for most of the casual labor employment in the village ... In many villages there is open or tacit collusion of big employers in the labor market” (p.60).

Further, under perfect competition, the introduction of a public workfare should result in an immediate increase in wages due to increased competition for labor by the government.¹⁴ This increase in private wages should be immediate (even in a multi-period environment) as there is complete upward adjustment of wages since the employers’ benefit of raising wages now outweigh the expected decline in future profits under perfect competition (see for e.g., [Kaur, 2019](#) for a discussion on employers’ response to positive shocks under perfect competition). However, the recent empirical literature on NREGA (see for e.g., [Berg et al., 2018](#); [Shrivastava, 2015](#); [Bahal, 2016](#)) does not find any immediate adjustment of wages due to the introduction of NREGA. This provides evidence of forward-looking employers having market power and withholding immediate wage increases, as against perfectly competitive labor markets.

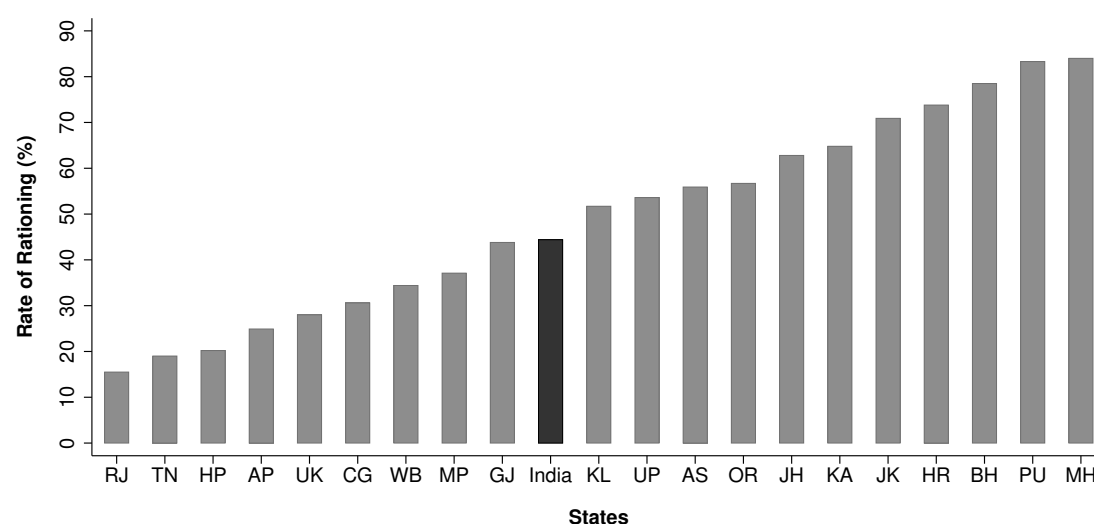
¹³National Rural Employment Guarantee Act (NREGA) is the largest public workfare in the world that was implemented by the Government of India in 2006. NREGA guarantees 100 days of employment at minimum wage to any household in rural India.

¹⁴In a competitive setting, large employment schemes can increase private wages in order to clear the labor supply and demand ([Imbert and Papp, 2015](#)).

4.2 Rationing in employment programs

Although NREGA guarantees 100 days of manual labor work to every rural household, work under NREGA is substantially rationed. Using evidence from the nationally representative National Sample Survey (NSS), [Dutta et al. \(2012\)](#) estimate rationing rates – defined as the proportion of workers who demand but do not get work under NREGA – for all the states of the country. **Figure 5** shows the rationing rates across Indian states calculated using estimates from [Dutta et al. \(2012\)](#). As the figure shows, while there is significant variation across states, rationing of work under NREGA is substantial with a national average rate of rationing of around 44% .

Figure 5: Rationing rates in NREGA across Indian states



Note: The rate of rationing is the proportion of workers who demand but do not get work under NREGA.
Source: [Dutta et al. \(2012\)](#).

[Muralidharan et al. \(2016\)](#) in their survey in the state of Orissa note that “less than 4 percent of workers in our control group report that they can access NREGA work whenever they want it.” Using data from a RICE survey conducted for 70 villages in Gujarat, Rajasthan, and Madhya Pradesh, [Imbert and Papp \(2020\)](#) also show that households reported only 38 days of work under NREGA. Moreover, not all the demand for work is even documented. As a World Bank study notes, many job card holders do not actively demand work and instead passively wait for work to be provided ([The World Bank, 2011](#)). Overall, there is conclusive evidence that work under such employment schemes is rationed.

If wages are fixed at a socially acceptable level and the budget for employment guarantees is limited, then the amount of employment provided under the program has to be adjusted accordingly. However, since both supply and demand for work under

the public workfare determine the level of program provision, it is important to check whether the variation in program provision is due to the differences in the demand for employment under public workfare. The evidence from recent studies, however, suggests that variation in employment provision is mostly due to supply factors like administrative capabilities and not from demand factors like local economic conditions or the level of poverty.

Imbert and Papp (2020) for example show that a host of district and worker level characteristics are unable to account for the stark differences in the level of employment provision across districts.¹⁵ Similarly, Dutta et al. (2012) find the number of NREGA days provided to be only weakly correlated with the level of poverty. Hence, there is evidence to support that fluctuations in program provision are mostly driven by supply-side factors like bureaucratic idiosyncrasies, political will, or experience in managing similar programs in the past. Most importantly, if variations in employment provision truly reflect demand for the program, then variability in program provision should not lead to compression of wage increases (as discussed in section 2). We empirically test the predictions of our model in section 5.

4.3 Variable Employment Guarantee

In this section, we show that the program provision in the future can deteriorate vis-à-vis its level today. The existence of such temporal variability in program provision is important since the compression of wage increases in our model is associated with the need to cut (downwardly rigid) wages in the future because of a decline in program provision. Using employment expenditure data from Bahal (2016), we use the *funds made available* at the start of each fiscal year as a measure of the level of program provision.¹⁶ Figure 6 shows the distribution of year-on-year growth of funds made available at the district level for two recent employment guarantee schemes in India between 2001 and 2010.

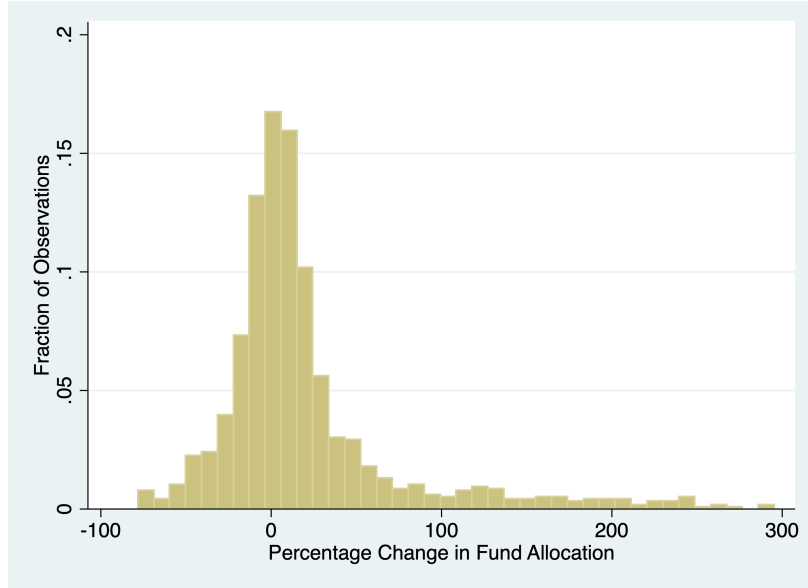
As can be seen, there are substantial fluctuations in the growth rates of fund allocation. While the distribution is skewed towards the right with an average of 16% growth, a standard deviation of 50% indicates a sizeable probability with which program provision may decline in the future. This is reaffirmed from the substantial mass of the distribution that lies to the left of zero.

More than one-third of the observations correspond to negative growth in fund al-

¹⁵Their district controls include for e.g., the rate of literacy, poverty rate, agricultural productivity, the level of wages, local elections, etc. Worker controls include age, sex, education, marital status, etc.

¹⁶Given that program provision is largely noted to be supply driven, funds made available at the start of a fiscal year are a good indicator of the intended level of program provision at the district level.

Figure 6: Distribution of Fund Allocation Changes



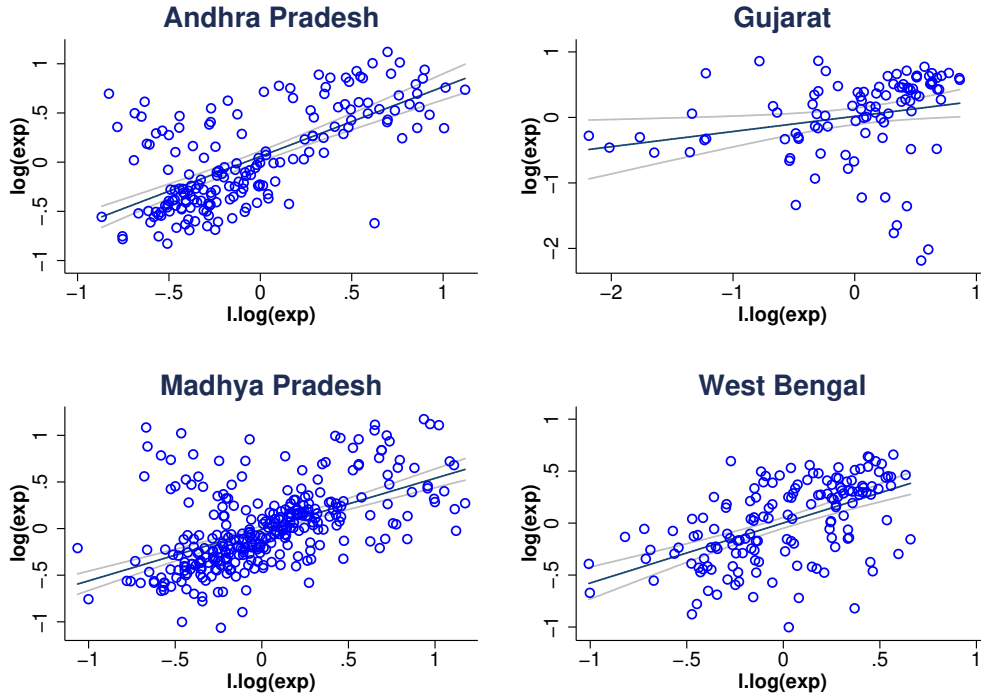
Note: The graph shows the distribution of year-on-year percentage change in real per capita fund allocation made under SGRY and NREGA. The unit of observation is a district-year. Data is for 442 districts from 2001-2010 (4420 observations). Fund allocation changes are top coded at 300%.

location (figure A3 plots the same graph separately for NREGA and SGRY. Both distributions have a substantial mass to the left of zero). It has been shown in the literature that there is substantial heterogeneity in rationing rates across the country (Dutta et al., 2012; Liu and Barrett, 2013). We find that there is also considerable heterogeneity in the probability of a decline in fund allocation across states, from just 15% for Andhra Pradesh to nearly 50% for Haryana. Furthermore, there is considerable temporal heterogeneity within a district regarding the growth of fund allocation and, as a consequence, the extent of rationing under the program. For example, the percentage of households that sought but did not receive work under NREGA in Medak district of Andhra Pradesh was 43% in 2007 but 21% in 2009 (Ravi and Engler, 2015). However, overall, there is a reasonable likelihood of the program provision deteriorating in the future compared to its current level.

Further, there is a high level of uncertainty associated with forecasting the level of program provision in the future. We show this using actual program expenditure data at the district level from 2001 to 2010 for SGRY and NREGA from Bahal (2016).¹⁷ Figure 7 plots the lead versus the current program expenditure (in logs) for 109 districts of four major states of India. As can be seen, actual expenditure in future can be sub-

¹⁷Sampoorna Grameen Rozgar Yojana (SGRY) was a nationwide public workfare that preceded NREGA and was launched in 2001. For details see Bahal (2016).

Figure 7: Predicted versus actual expenditure using past expenditure



stantially higher or lower than the predicted 95% confidence interval obtained from a linear regression of future expenditure $\ln(e_{i,t+1})$ on current expenditure $\ln(e_{i,t})$.¹⁸ This is evidence of the uncertainty that exists regarding the level of program implementation in the future. Note that this figure is illustrative only, and that the uncertainty could be because of any number of district-year specific covariates.

4.4 Downward Nominal Wage Rigidity

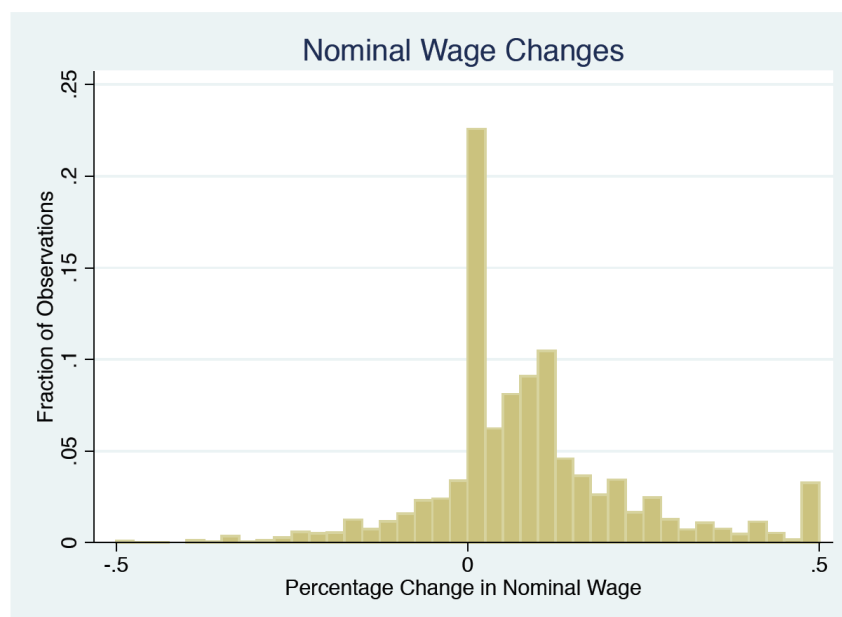
Finally, our assumption of downward nominal wage rigidity follows from the evidence presented in Kaur (2019) regarding year-on-year changes in nominal agricultural wages in India. Figure 8 (from Kaur, 2019) shows nominal wage changes for 256 districts over a period of 30 years.

The distribution of wage changes exhibits large clustering at zero and a discontinuous drop to the left of zero. This may be indicative of potential wage cuts that did not materialize due to downward rigidities.¹⁹ Kaur further concludes that while posit-

¹⁸The regression also controls for district and year fixed effects along with a binary variable $Imp_{i,t}$ which indicates whether or not NREGA was implemented in a district by year t .

¹⁹The large cluster at nominal zero changes is not expected under a continuous distribution of shocks (Kahn, 1997). In contrast, Kaur (2019) shows the distribution of real agricultural wage changes to be

Figure 8: Changes in nominal wages



Note: The figure is a histogram of year-on-year percentage changes in nominal agricultural wages for 256 districts from 1956-1987 (7680 obs.). Wage changes are top coded at 50% and bottom coded at -50%.
Source: Kaur (2019), World Bank Climate and Agriculture dataset.

ive rainfall shocks increase nominal wages, negative shocks do not result in wage cuts. Such asymmetric wage adjustment to labor demand shocks is consistent with downward rigidities.

5 Application to employment programs in India: Testing the Theory

Our model generates two testable predictions: (i) that average positive marginal effect of the workfare on private wages should be lower when variation in program provision is high; and (ii) that the compression of wage increases should be more severe when inflation is low relative to when inflation is high. The first prediction follows from section 2.3 where we show that greater variability in program provision results in a larger compression of wage increases (figure 4). Next, as is discussed in section 3.1, price inflation makes the constraint of downward nominal wage rigidity less restrictive. This is so since relative to the constant price assumption, price inflation alleviates the need for future wage cuts (due to a decline in program provision) resulting in a smaller

symmetric around zero. Survey evidence from Kaur further suggests that nominal wage cuts are regarded as unfair by agricultural workers and employers.

compression of wage increases today.

To empirically validate the predictions of the model, we use district-level expenditure data from 2001-2010 for two large public workfare programs in India: SGRY and NREGA as used in Bahal (2016).²⁰ The annual frequency data is reported in financial year format (1 April to 31 March) and includes information on the opening balance, funds made available at the start of the fiscal year, and expenditure incurred over the year. While data on SGRY is from the year of its implementation in 2001 to its last operational financial year 2007-08, data on NREGA starts from 2006-07 to 2010-11. Next, we use district-level wage data on agricultural activities as reported by the Agricultural Wages of India (AWI). The monthly frequency AWI data is from July 2001 - June 2011. We use the daily wage rate data for *field labor* which includes agricultural activities like ploughing, sowing, reaping, and weeding. Since wage rates are reported separately for men and women, we take the average of male and female wages to construct our measure of field wages.²¹

A shortcoming of the monthly AWI wage series is that it often contains missing data for some of the months. Furthermore, the annual publications of AWI sporadically exclude data for some districts and states.²² We first convert the monthly wage series to annual frequency by taking 12-month averages in the financial year format to match the frequency and the period of the employment expenditure data. Second, we restrict our attention to a complete balanced panel of 134 districts from 12 major states of India. This is required for the construction of our measure of variability, as we explain in the next subsection. Hence, we have a dataset of 1340 annual observations. We deflate both the wage and employment expenditure data to 2001 prices using state-wise Consumer Price Index for Rural Laborers (CPI-RL) collected by the Labor Bureau, Government of India. Unless otherwise mentioned, all the variables in the empirical analysis are in real, per-capita terms, considering only the rural populations of the respective districts. Summary statistics for the main variables are given the Appendix.

5.1 Empirical specification

To test the two predictions of our theory, we estimate equations 3 and 4 where the subscripts i and t denote district and year respectively. The dependent variable is agricultural wages $w_{i,t}$ (in rupees), while expenditure under the program $e_{i,t}$ is a measure

²⁰The data is collected from district-level financial statements of SGRY and NREGA from (i) Ministry of Rural Development (MoRD); (ii) Datanet (India); and (iii) nrega.nic.in.

²¹Although AWI reports wages for children as well, we exclude wages reported under children as most of the observations are missing under this category.

²²For e.g., data for states like Chhattisgarh, Jharkhand, and Uttarakhand is not available before 2005.

of program provision (also in rupees). Ex-ante, we expect wages to increase as expenditure under the program increases. For a district, we represent the expected future variability in program provision in district i in time t by $\sigma_{i,t}$.

$$w_{i,t} = \beta_1 e_{i,t} + \beta_2 \sigma_{i,t} + \beta_3 (e_{i,t} \times \sigma_{i,t}) + \alpha_i + \gamma_t + \xi_{it} + \beta \mathbf{X}_{i,t} + Overspend_{i,t} + \varepsilon_{i,t} \quad (3)$$

$$w_{i,t} = \beta_1 e_{i,t} + \beta_2 \sigma_{i,t} + \beta_3 (e_{i,t} \times \sigma_{i,t}) + \beta_4 (e_{i,t} \times \sigma_{i,t} \times Low\ inflation_i) + \alpha_i + \gamma_t + \xi_{it} + \beta \tilde{\mathbf{X}}_{i,t} + Overspend_{i,t} + \varepsilon_{i,t} \quad (4)$$

According to our model, forward looking employers take into account future variability of program provision. We assume that employers form expectations about future variability using past and present realizations of expenditure. In other words, the information set of employers in the private labor market is limited to current and past levels of program provision (expenditure) and this information set is continuously updated every period. Given the data, we construct a measure of variability, $\sigma_{i,t}$, which is the standard deviation of program expenditure in district i and year t using expenditure till year t : $\{e_{i,2001}, \dots, e_{i,t}\}$ that is calculated recursively for $t = 2002, \dots, 2010$.

Since greater variability in program provision is expected to compress wage increases, we expect the marginal effect of expenditure on wages to decrease as $\sigma_{i,t}$ increases. That is, we expect the coefficient of interaction β_3 to be negative.

To test the second prediction, we use equation 4 to test whether the rate of inflation has any influence on the degree of compression. We calculate annual inflation using Consumer Price Index-Rural Laborer (CPI-RL) for each district over the ten-year period between 2001 and 2010. We then divide the sample of 134 districts between low inflation and high inflation districts based on whether a district is respectively below or above the median value of annual inflation averaged over the ten-year period (the median is 6.8%). The variable *Low inflation_i* is a dummy variable indicating the district being low inflation.²³ As the wage compression is predicted to be higher in low inflation districts, we would expect the coefficient β_4 to be negative.

Among other controls, α_i and γ_t represent district and year fixed effects respectively; ξ_{it} controls for district specific trends; $Imp_{i,t}$ signifies a change in program regime for a district from SGRY to NREGA. Finally, $\mathbf{X}_{i,t}$ is a vector of additional controls that may simultaneously affect wages and program provision in a district. These include an indicator variable for the regime change from SGRY to NREGA, an indicator for state

²³In theory, what matters is the expected inflation in the next period. We use the average actual inflation over 10 years to classify districts as high or low inflation. The assumption here is that districts that have high average actual inflation would also have high expected inflation.

election-year, the proportion of Scheduled Castes and Scheduled Tribes population in a district, the average rainfall (in millimeters) that a district receives in a year during the rainy season, and lagged program expenditure.²⁴ $\tilde{\mathbf{X}}_{i,t}$ in equation 4 includes all controls that are in $\mathbf{X}_{i,t}$ along with all relevant double interaction terms. Table A1 provides summary statistics of the key variables used in empirical analysis. Since the district sizes vary in our panel, we weight all the regressions by district population. Further, we relax the assumption that within district observations are independent by clustering standard errors at the district level that are robust to heteroskedasticity as well.

While the specification in equation 3 controls for a host of endogeneity concerns, it is still susceptible to district-year fluctuations in program expenditure that can be arguably endogenous. For example, any local shock like natural calamity or conflict may adversely affect the local agricultural labor market and result in higher than expected program expenditure due to an increase in demand for public work.²⁵

Following Bahal (2016), we use the fund allocation process of the programs to check for potentially endogenous program expenditure. Due to the structure of fiscal federalism in India, virtually all of the expenditure under such programs is financed by the central government. This entails a certain amount of funds that are made available at the start of the fiscal year for all the districts. However, expenditure during the whole year is not necessarily restricted by the fund availability since the financial accounts roll over to the next year. This allows actual expenditure to exceed fund availability (at district level) in case the demand for work under the program is unusually high, and vice versa. If there is over-expenditure, the central government clears the negative opening balance next year by making the appropriate releases to (i) meet previous obligations and (ii) make funds available for the following year.

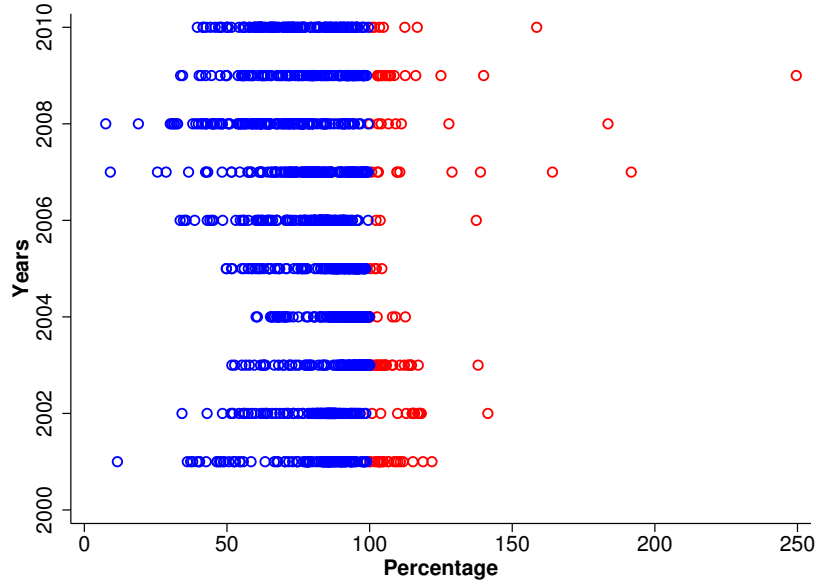
Figure 9 plots the year-wise utilization of funds for 134 districts. The observations marked in red show over-utilization while the observations in blue represent under-utilization. If the under/over utilization of funds is related to the demand for work under the program due to private labor market conditions, then this may result in biased OLS estimates of equation 3. To control for such potentially endogenous district-year fluctuations, we add the control variable $Overspend_{i,t}$ which is the percentage over-utilization (under-utilization will be negative) of the funds allocated.

Even after controlling for endogenous expenditure changes in this manner, there could still be some residual endogeneity. In subsection 5.3, we conduct a battery of

²⁴We use remote sensed rainfall data from the Tropical Rainfall Measuring Mission (TRMM) satellite. See Fetzer (2020) for a detailed discussion on the consistency and the quality of TRMM data over any other remote sensed or ground-based data. We thank Thiemo Fetzer for sharing the rainfall data.

²⁵Drèze (1990) for example discusses a high take up of public employment by laborers under Maharashtra's employment guarantee scheme during the famine of 1970-73.

Figure 9: Actual Expenditure as a Percent of Fund Availability



Note: The figure shows the year-wise utilization of funds as a percent of funds made available. Observations marked in red (blue) show over (under) utilization.

Source: Bahal (2016).

endogeneity checks, including the Oster (2019) test, to show that our baseline estimates are unlikely to suffer from any consequential bias.²⁶

5.2 Results

Tables 1 and 2 report results based on equations 3 and 4, respectively. In table 1, columns 1, 2, 3, and 4 progressively add more controls where column 4 is our preferred specification. We interpret the economic significance of the coefficient estimates from the full specification of equation 3 in column 4.

The coefficient of 2.406×10^{-2} for $e_{i,t}$ suggests a wage increase of 6.42% (of the average wage) computed at mean expenditure $\bar{e}$ and $\sigma_i = 0$.²⁷ This estimated increase in wages closely matches the wage increase reported by (Berg et al., 2018; Imbert and Papp, 2015). Importantly, the coefficient of the interaction term $e_{i,t} \times \sigma_{i,t}$ is negative and statistically significant. We compute the marginal effect of expenditure on wages at the average ($\bar{\sigma}_{i,t} \approx 50.38$ rupees). We find that relative to when $\sigma_{i,t} = 0$, the marginal effect of expenditure on wages contracts by approx. 12.6% ($0.006 \times 50.38 / 2.406 = 0.126$) at

²⁶The Oster (2019) test is based on the idea that, given the effect of observable controls on the coefficient of interest, we can estimate the degree of correlation that unobservables would need to have so that the unbiased coefficient is actually zero (Altonji et al., 2005).

²⁷The average per-capita expenditure is ₹152 while the average wage is around ₹57 (2001 prices).

Table 1: Effect of Variability in Expenditure on Wages

	(1)	(2)	(3)	(4)
$e_{i,t}$	2.518 [0.832]	3.273 [0.927]	2.096 [0.740]	2.406 [0.751]
$\sigma_{i,t}$	1.838 [2.005]	-0.524 [2.121]	-2.913 [2.609]	-3.142 [2.664]
$e_{i,t} \times \sigma_{i,t}$	-0.007 [0.002]	-0.009 [0.003]	-0.005 [0.002]	-0.006 [0.002]
Fixed effects	Yes	Yes	Yes	Yes
Other Controls	No	Yes	Yes	Yes
District time trends	No	No	Yes	Yes
$Overspend_{i,t}$	No	No	No	Yes
Observations	1206	1206	1206	1206
Adjusted R^2	0.896	0.898	0.946	0.946

All estimates in the table are of the order of magnitude 10^{-2} . The unit of observation is a district-year. Data is for 134 districts from 2001-2010. The dependent variable in all the regressions is real agricultural wage. $e_{i,t}$ is real per-capita expenditure. For a district, $\sigma_{i,t}$ is the standard deviation calculated till year t $\{e_{i,2001}, \dots, e_{i,t}\}$ for $t = 2002, \dots, 2010$. $\pi_{i,t}$ is an indicator that takes the value 1 for low inflation districts and zero otherwise. Fixed effects include district and year fixed effects together with district specific heterogeneous trends. Other controls include an indicator for a change in program regime $Imp_{i,t}$, an indicator for state election-year, the proportion of scheduled castes and scheduled tribes population in a district, the average rainfall (in millimeters) that a district receives in a year during the rainy season, and lagged expenditure. $Overspend_{i,t}$ is the difference between actual expenditure and fund availability taken as a percent of fund availability. Standard errors are clustered at the district level and are reported in parentheses.

Table 2: Differential Effect across High and Low Inflation Districts

	(1)	(2)	(3)	(4)	(5)
$e_{i,t}$	2.292 [1.022]	3.140 [1.105]	2.154 [0.782]	2.519 [0.790]	2.191 [0.776]
$\sigma_{i,t}$	2.139 [2.037]	-0.450 [2.156]	-1.821 [2.746]	-2.107 [2.802]	
$e_{i,t} \times \sigma_{i,t}$	-0.007 [0.003]	-0.008 [0.003]	-0.006 [0.003]	-0.007 [0.003]	-0.007 [0.002]
$e_{i,t} \times \sigma_{i,t} \times \pi_i$ (Low)	-0.036 [0.026]	-0.034 [0.025]	-0.010 [0.023]	-0.010 [0.023]	-0.011 [0.005]
$e_{i,t} \times \pi_i$ (Low)	3.894 [3.392]	3.495 [3.314]	1.652 [2.615]	1.630 [2.584]	
$\sigma_{i,t} \times \pi_i$ (Low)	6.473 [5.428]	7.026 [5.335]	-6.687 [6.801]	-6.544 [6.818]	
Fixed effects	Yes	Yes	Yes	Yes	Yes
Other Controls	No	Yes	Yes	Yes	Yes
District time trends	No	No	Yes	Yes	Yes
$Overspend_{i,t}$	No	No	No	Yes	Yes
Observations	1206	1206	1206	1206	1206
Adjusted R^2	0.896	0.898	0.946	0.946	0.946

All estimates in the table are of the order of magnitude 10^{-2} . The unit of observation is a district-year. Data is for 134 districts from 2001-2010. The dependent variable in all the regressions is real agricultural wage. $e_{i,t}$ is real per-capita expenditure. For a district, $\sigma_{i,t}$ is the standard deviation calculated till year $t \{e_{i,2001}, \dots, e_{i,t}\}$ for $t = 2002, \dots, 2010$. $\pi_{i,t}$ is an indicator that takes the value 1 for low inflation districts and zero otherwise. Fixed effects include district and year fixed effects together with district specific heterogeneous trends. Other controls include an indicator for a change in program regime $Imp_{i,t}$, an indicator for state election-year, the proportion of scheduled castes and scheduled tribes population in a district, the average rainfall (in millimeters) that a district receives in a year during the rainy season, and lagged expenditure. $Overspend_{i,t}$ is the difference between actual expenditure and fund availability taken as a percent of fund availability. Standard errors are clustered at the district level and are reported in parentheses.

$\bar{\sigma}_{i,t}$. Finally, consistent with the hypothesis that variability in program provision should have no impact on wages at zero expenditure, the coefficient of $\sigma_{i,t}$ is insignificantly different from zero (since $\frac{\partial w}{\partial \sigma} = \beta_2 + \beta_3 e$ and $\frac{\partial w}{\partial \sigma}|_{e=0} = 0 \Rightarrow \beta_2 = 0$).

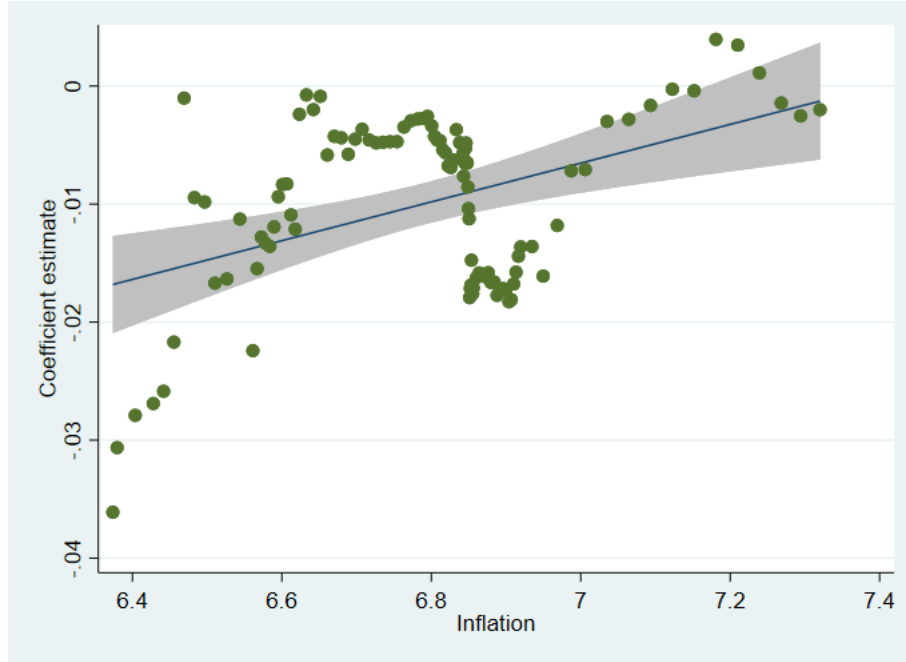
In table 2, columns 1, 2, 3 and 4 report results for equation 4 including the complete set of interaction terms with progressively more controls added. The variable of interest is $e_{i,t} \times \sigma_{i,t} \times \pi_i$ (*Low*) whose coefficient is negative as expected but is statistically insignificantly different from zero. This could be because our inflation data is at the state-level and hence all districts in a state are clubbed together either in the high or the low inflation category. This introduces measurement error that may attenuate the coefficient towards 0. Another possibility is the problem of multicollinearity emanating from redundant regressors and other interaction terms, which could be inflating standard errors. Since the coefficients of $\sigma_{i,t}$, $e_{i,t} \times \pi_i$ (*Low*), and $\sigma_{i,t} \times \pi_i$ (*Low*) in column 4 are not jointly significant (F-stat 1.55, p-value 0.21), column 5 reports the results of estimating equation 4 without these regressors. As can be seen, while the coefficient of $e_{i,t} \times \sigma_{i,t} \times \pi_i$ (*Low*) remains the largely the same, the standard errors reduce by a factor of five indicating that the exclusion of the regressors does not bias the coefficient estimate but yields a more precise estimate. The results noted in column 4 imply that the compression in low inflation districts is approximately 25 percentage points more than high inflation districts.

The above exercise provides suggestive evidence in favor of our second prediction but cannot conclusively show that high inflation districts have lower wage compression. To show this, we perform rolling regressions of 30 districts (300 observations) at a time, starting from those with the lowest inflation and moving the window up one district at a time through the list of districts sorted by average inflation, till we reach the 30 districts with the highest average inflation. We run our full specification of equation 3 and record the estimate of β_3 , which is the coefficient of $e_{i,t} \times \sigma_{i,t}$. We also record the mean inflation of the set of 30 districts. In figure 10, we plot the coefficient estimates versus the mean inflation. We see a clear upward slope showing that the wage compression, indicated by the coefficient estimate's magnitude, is much higher in low inflation districts and moves close to 0 in districts with high inflation. This provides strong support for the second prediction of our model.

Hence, in agreement with studies like [Elsby \(2009\)](#) and [Stüber and Beissinger \(2010\)](#), we find compression of wage increases to be larger when inflation is low. This suggests that while other channels like risk aversion may potentially cause wage compression as well, the nominal wage rigidity channel that we describe is an important, if not the only channel, causing wage compression.

The above results further corroborate the central message of the paper, incomplete

Figure 10: Coefficients from rolling regressions



Note: Each point in the graph denotes a set of 30 districts (300 district-year observations). The vertical axis is the coefficient of $e_{i,t} \times \sigma_{i,t}$ obtained from estimating equation 3 over these observations. The horizontal axis is the mean value of the average inflation rate in these districts.

and variable implementation of public workfares can result in compression of wage increases in the presence of downward nominal wage rigidity. Further, the degree of compression is more severe when inflation is low than when inflation is high.²⁸

One possible objection to our claim can be that districts where program provision is highly variable could also suffer from high levels of corruption which can explain the attenuation of wage increases. The existence of significant levels of corruption and its dynamics have been well documented (Niehaus and Sukhtankar, 2013a,b). The effect of corruption on labor market outcomes including private wages has also been demonstrated (Muralidharan et al., 2016, 2018). However, if corruption alone leads to a compression of wage increases, then the degree of compression should not be sensitive to inflation. On the other hand, if variability in program provision and nominal wage rigidity are important mechanisms through which compression of wage increases occurs, then areas with higher rates of inflation should have a lower compression of wage increases, which is what we find.²⁹

²⁸It is important to acknowledge that the implementation of a large public workfares like NREGA can itself have general equilibrium effects on the aggregate price level. We, however, do not find any association of NREGA implementation with high inflation after controlling for district and year fixed effects. Results available on request.

²⁹We also show results with nominal wage as the dependent variable in the appendix. The results

5.3 Endogeneity checks

Despite the extensive set of controls and fixed effects we use, it is possible that there may be omitted variables that may be correlated with either the level or the variability of expenditure, leading to biased coefficients. While we are not able to completely rule out this possibility, we perform a few checks to show that endogeneity may not be a serious cause for concern here.

5.3.1 Rainfall

While we control for district and year fixed effects, and district specific time trends, there may be district-year specific shocks that may affect both expenditure and wages and hence bias the coefficients. One such shock could be rainfall, and hence we control for it in our preferred specification. Here we check if rainfall predicts the level or variability of expenditure. The results are shown in tables 3 and 4. We find no correlation between rainfall and the level or variability of expenditure, hence alleviating to some extent the concern that district-year specific shocks could be biasing the results.

Table 3: Orthogonality of $e_{i,t}$ with rainfall

	(1)	(2)	(3)	(4)	(5)
$Rain_{i,t}$	-0.015 [0.046]	0.049 [0.057]	0.062 [0.053]	0.008 [0.051]	0.019 [0.051]
Fixed effects	No	Yes	Yes	Yes	Yes
Other Controls	No	No	Yes	Yes	Yes
District time trends	No	No	No	Yes	Yes
$Overspend_{i,t}$	No	No	No	No	Yes
Observations	1340	1340	1206	1206	1206
Adjusted R^2	-0.001	0.559	0.773	0.801	0.817

The dependent variable in all the regressions is $e_{i,t}$, real per-capita expenditure. $Rain_{i,t}$ is the average rainfall (in millimeters) that a district receives in a year during the rainy season.

5.3.2 Fund allocation

To further alleviate endogeneity concerns, we check if the results hold if we replace actual program expenditure with the expected outlays or funds that the central govern-
remain similar.

Table 4: Orthogonality of $\sigma_{i,t}$ with rainfall

	(1)	(2)	(3)	(4)	(5)
$Rain_{i,t}$	-0.032 [0.020]	0.036 [0.024]	0.033 [0.017]	0.013 [0.015]	0.016 [0.015]
Fixed effects	No	Yes	Yes	Yes	Yes
Other Controls	No	No	Yes	Yes	Yes
District time trends	No	No	No	Yes	Yes
$Overspend_{i,t}$	No	No	No	No	Yes
Observations	1206	1206	1206	1206	1206
Adjusted R^2	0.002	0.589	0.879	0.921	0.925

The dependent variable in all the regressions is $\sigma_{i,t}$, variability in real per-capita expenditure. $Rain_{i,t}$ is the average rainfall (in millimeters) that a district receives in a year during the rainy season.

ment allocates to each district at the start of a financial year. While the actual expenditure may be affected by district-year specific unobservable shocks, funds allocated at the beginning of the financial year are likely to be independent of these shocks. In table 5, we replace actual expenditure $e_{i,t}$ and variability in program expenditure $\sigma_{i,t}$ with fund availability $e_{i,t}^a$ and variability in fund availability $\sigma_{i,t}^a$, respectively. Given the coefficient of 2.24×10^{-2} and the average fund availability of 195.32, column 4 suggests a wage increase of 7.68% at $\sigma_{i,t}^a = 0$, very comparable to 6.42% reported for the baseline specification reported in table 1. The coefficient of the interaction term -0.003 suggests a wage compression of 8.5% at mean $\bar{\sigma}_{i,t}^a = 63.47$ relative to $\sigma_{i,t}^a = 0$, again very comparable to 12.6% obtained using actual expenditure.

5.3.3 Oster Test

Following Oster (2019), table 6 presents the endogeneity tests' results for the coefficients of interest in columns 4 and 5 of tables 1 and 2, respectively. The test estimates the ratio (δ) of the correlation between the variable of interest and potentially unobserved omitted variable, to that between the variable and observables, that would be required to reduce the coefficient to zero. For all coefficients this ratio indicates that the correlation with unobservables needs to be at least an order of magnitude higher to negate the effect.

Further, given δ we estimate an identified set of the true effect $[\beta_{controlled}, \beta(\delta, R_{max})]$. Following Oster (2019), we regress wage on its lag and other controls to establish a

Table 5: Results with outlay instead of expenditure

	(1)	(2)	(3)	(4)
$e_{i,t}^a$	2.655 [0.608]	3.415 [0.734]	2.282 [0.606]	2.242 [0.612]
$\sigma_{i,t}^a$	-0.403 [1.451]	-2.810 [1.774]	-4.181 [2.431]	-4.108 [2.421]
$e_{i,t}^a \times \sigma_{i,t}^a$	-0.004 [0.001]	-0.005 [0.001]	-0.003 [0.001]	-0.003 [0.001]
Fixed effects	Yes	Yes	Yes	Yes
Other Controls	No	Yes	Yes	Yes
District time trends	No	No	Yes	Yes
$Overspend_{i,t}$	No	No	No	Yes
Observations	1206	1206	1206	1206
Adjusted R^2	0.897	0.899	0.946	0.946

All estimates in the table are of the order of magnitude 10^{-2} . The table reports the same set of regressions as in table 1 but after replacing actual expenditure and volatility in actual expenditure with fund availability and volatility in fund availability, respectively.

Table 6: Oster Test

Variable	Column 4, table 1		Column 5, table 2	
	$[\beta_c, \beta(\delta, R_{max})]$	$\delta _{\beta=0}$	$[\beta_c, \beta(\delta, R_{max})]$	$\delta _{\beta=0}$
$e_{i,t}$	[2.406, 2.599]	-23.96	[2.191, 2.326]	-28.08
$e_{i,t} \times \sigma_{i,t}$	[-0.006, -0.006]	36.41	[-0.007, -0.007]	40.84
$e_{i,t} \times \sigma_{i,t} \times \pi_{i,t}$			[-0.011, -0.011]	-810.53

The table reports the identified set that contains the true coefficient of the relevant variables reported in columns 4 and 5 of tables 1 and 2, respectively. β_c is the elasticity estimate as reported in tables 1 and 2 while $\beta(\delta, R_{max})$ is computed at $\delta = 1$ and $R_{max} = 0.96$. β_c and $\beta(\delta, R_{max})$ are of the order of magnitude 10^{-2} . The table also reports the coefficient of proportionality that is required to explain away the treatment effect after adding the unobservables in a hypothetical regression.

theoretical R_{max} of 0.96. Again, all intervals do not include zero, thus making it very likely that even in the presence of some residual endogeneity, the results we obtained in

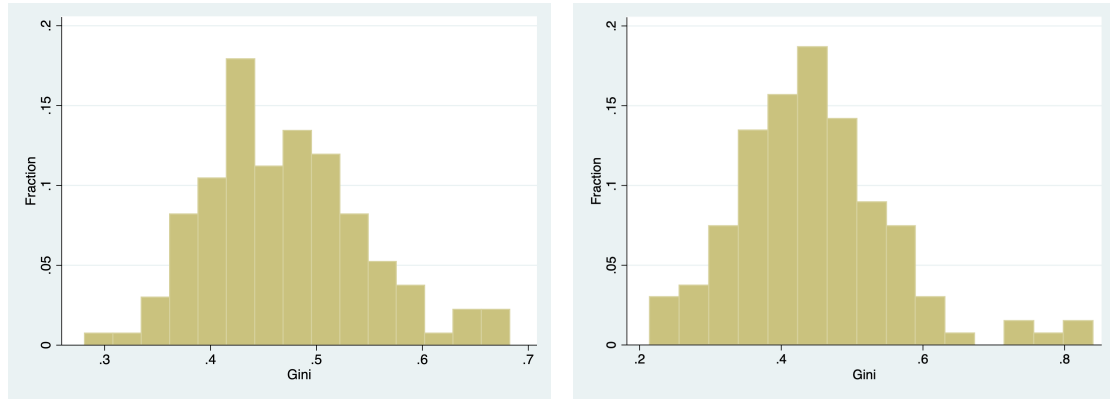
columns 4 and 5 of tables 1 and 2 remain robust, respectively.

5.4 Robustness checks

5.4.1 Robustness to the degree of market power

As described in section 3.2, we expect to see compression of wage increases as a result of variability in program expenditure, regardless of the degree of employers' market power. To check this we use district level data on concentration of agricultural land. We calculate the district level Gini of land cultivated in rural areas using household survey data from two rounds (2004-05 and 2011-12) of the National Sample Survey. Figure 11 shows the distribution of the Ginis, indicating a rather unequal distribution of land ownership in these districts.

Figure 11: Land concentration



(a) Distribution of Land Gini from 2004-05

(b) Distribution of Land Gini from 2011-12

Note: Figure 11(a) and Figure 11(b) show the distribution of district level Gini coefficients for land cultivated by households in rural areas using data from the National Sample Survey from 2004-05 and 2011-12 respectively

If we assume higher land concentration to indicate the greater degree of employer market power, then our prediction from subsection 3.2 suggests that there should be no significant difference in the degree of wage compression between districts with high and low land concentration. We test this using the following specification:

$$w_{i,t} = \beta_1 e_{i,t} + \beta_2 \sigma_{i,t} + \beta_3 (e_{i,t} \times \sigma_{i,t}) + \beta_4 (e_{i,t} \times \sigma_{i,t} \times Low\ concentration_i) + \alpha_i + \gamma_t + \xi_{i,t} + \beta \hat{X}_{i,t} + Overspend_{i,t} + \varepsilon_{i,t} \quad (5)$$

Here, *Low concentration_i* is an indicator variable that takes the value 1 for districts where the Gini coefficient of agricultural land cultivated is below the median. All other

variables have the same interpretation as equation 3 and $\dot{\mathbf{X}}_{i,t}$, in addition, includes all relevant double interaction terms as well. As discussed earlier, if wage compression is independent of the degree of competition in the labor market, β_4 should be close to zero. Table 7 confirms this from both sets of data. Column 1 is the result from our preferred specification. Columns 2 and 3 show the results of interactions with the

Table 7: Robustness with varying market power

	(1)	(2)	(3)
$e_{i,t}$	2.406 [0.751]	2.730 [1.001]	2.821 [1.191]
$\sigma_{i,t}$	-3.142 [2.664]	-3.173 [2.691]	-4.476 [3.313]
$e_{i,t} \times \sigma_{i,t}$	-0.006 [0.002]	-0.006 [0.003]	-0.005 [0.004]
$e_{i,t} \times \sigma_{i,t} \times \text{Conc.}(\text{Low})$		0.002 [0.005]	-0.001 [0.005]
$e_{i,t} \times \text{Conc.}(\text{Low})$		-0.670 [1.291]	-0.664 [1.350]
$\sigma_{i,t} \times \text{Conc.}(\text{Low})$		-0.084 [3.912]	1.920 [3.709]
All controls	Yes	Yes	Yes
Observations	1206	1206	1206
Adjusted R^2	0.946	0.946	0.946

All estimates are of the order 10^{-2} . Column 1 is the preferred specification reported in column 4 of table 1. Column 2 reports results obtained by estimating equation 5 using land concentration data from National Sample Survey (NSS) round 61 (2004-2005). Column 3 reports results for the same specification but using land concentration data from NSS round 68 (2011-2012). Standard errors are clustered at the district level and are reported in parentheses.

Low concentration variable defined above, from National Sample Survey data collected during 2004-05 and 2011-12, respectively. The estimate for β_4 is statistically and economically insignificant in both columns. Hence, to the extent that land concentration indicates market power, our results are robust to variations in it.³⁰

³⁰Further, if large farmers and employers view public workfares as adversely affecting them, they may

It is important to note that while the ratcheting mechanism discussed in [subsection 3.2](#) is a necessary condition for wage compression under perfect competition, it is also applicable, although to a lesser degree, when markets are monopsonistic. To test this mechanism empirically, what actually matters is the highest level of program implementation (or the lowest rate of rationing) in the past. If wage increases are compressed due to the inertia of high wages that employers inherit from the past, then we should observe wages to be less responsive to current program expenditure in regions where the maximum expenditure in the past has been high. We find that the data does not support this prediction (please see Appendix), suggesting that the ratcheting mechanism may not be the dominant mechanism through which variability in workfare expenditure compresses wage increases. This test, therefore, provides some suggestive, but by no means conclusive, evidence against the presence of perfect competition in agricultural labor markets in India.

5.4.2 Robustness to price volatility

It is important to acknowledge that apart from uncertainty regarding program expenditure, numerous other uncertainties exist which may prevent employers from increasing wages fully. For example, price fluctuations may impede employers' ability to increase wages freely in response to public workfares. If there is an expectation that future agricultural prices can drop, farmers may not want to raise wages in response to higher program expenditure now. If fluctuations in prices and program expenditure are correlated, there would be a bias in the results. To check for this, we estimate equation 3 with an additional covariate $\sigma_{i,t}^p$ in table 8. $\sigma_{i,t}^p$ is the standard deviation of CPI-RL index calculated analogously to $\sigma_{i,t}$. Column 1 reports the standard result obtained by estimating equation 3. Column 2 additionally controls for $\sigma_{i,t}^p$. Finally, column 3 also adds an interaction term $e_{i,t} \times \sigma_{i,t}^p$ to check if program expenditure affects wages differentially based on volatility in prices. The coefficient of $\sigma_{i,t}^p$ in columns 2 and 3 and $e_{i,t} \times \sigma_{i,t}^p$ in column 3 all have the expected negative sign. This suggests price volatility to have a negative effect on agricultural wages, both directly and also by attenuating the positive effect of program expenditure on wages. These coefficients, however, are not significantly different from zero. Encouragingly, the key coefficients of $e_{i,t}$ and $e_{i,t} \times \sigma_{i,t}$ in columns 2 and 3 remain very comparable to the baseline specification in column 1.

influence funding allocation or its variability. Encouragingly, using the landholding data, we find that land concentration is uncorrelated with both the level and variance of expenditure in our sample (Pearson's correlation coefficients <0.1) allaying any endogeneity concerns arising from such correlations.

Table 8: Controlling for volatility in prices

	(1)	(2)	(3)
$e_{i,t}$	2.406 [0.751]	2.302 [0.744]	2.858 [1.022]
$\sigma_{i,t}$	-3.142 [2.664]	-2.509 [2.535]	-2.951 [2.582]
$e_{i,t} \times \sigma_{i,t}$	-0.006 [0.002]	-0.005 [0.002]	-0.005 [0.002]
$\sigma_{i,t}^p$		-7.798 [4.842]	-6.891 [4.709]
$e_{i,t} \times \sigma_{i,t}^p$			-0.005 [0.004]
All controls	Yes	Yes	Yes
Observations	1206	1206	1206
Adjusted R^2	0.946	0.946	0.946

All estimates in the table are of the order of magnitude 10^{-2} . Column 1 reports the same regression results reported in column 4 of table 1. Column 2, in addition, controls for $\sigma_{i,t}^p$ which is the standard deviation of CPI-RL calculated till year t $\{e_{i,2001}, \dots, e_{i,t}\}$ for $t = 2002, \dots, 2010$. Column 3 also controls for an interaction term $e_{i,t} \times \sigma_{i,t}^p$.

5.4.3 Robustness to the seasonality of wages

Since the average wage over the whole year may also include the effect of events like short term out-migration during the non-agricultural/lean season, in table 9 we check if the results hold if we replace average agricultural wages over the whole year with average wages during the agricultural/peak season (Jul-Dec). On the other hand, the literature has also noted that most of the employment under the workfares may happen during the lean season as the government avoids competing with local farmers for laborers during the peak agricultural season. To see if the results are substantially different during the non-agricultural/lean season, we also show the results with average agricultural wages during the months Jan-Jun in table 10.

Table 9: Results with avg. wages in the peak season

	(1)	(2)	(3)	(4)
$e_{i,t}$	3.959 [0.803]	4.929 [0.881]	2.665 [0.844]	3.204 [0.861]
$\sigma_{i,t}$	-0.801 [2.238]	-4.739 [2.356]	-4.199 [2.880]	-4.564 [2.949]
$e_{i,t} \times \sigma_{i,t}$	-0.009 [0.002]	-0.011 [0.002]	-0.005 [0.003]	-0.006 [0.003]
Fixed effects	Yes	Yes	Yes	Yes
Other Controls	No	Yes	Yes	Yes
District time trends	No	No	Yes	Yes
$Overspend_{i,t}$	No	No	No	Yes
Observations	1093	1093	1093	1093
Adjusted R^2	0.900	0.902	0.939	0.939

All estimates in the table are of the order of magnitude 10^{-2} . The table reports the same set of regressions as table 1 but with the dependent variable as the average agricultural wages during the agricultural/peak season (Jul-Dec).

Overall, the results reported in tables 9 and 10 are consistent with the baseline results reported in table 1. The preferred specification in column 4 suggests a wage compression of 9.4% and 11.9% for tables 9 and 10, respectively.

Table 10: Results with avg. wages in the lean season

	(1)	(2)	(3)	(4)
$e_{i,t}$	2.719 [0.881]	3.450 [0.982]	2.259 [0.830]	2.539 [0.838]
$\sigma_{i,t}$	1.550 [2.119]	-0.372 [2.265]	-2.854 [2.785]	-3.085 [2.789]
$e_{i,t} \times \sigma_{i,t}$	-0.008 [0.002]	-0.009 [0.003]	-0.006 [0.003]	-0.006 [0.003]
Fixed effects	Yes	Yes	Yes	Yes
Other Controls	No	Yes	Yes	Yes
District time trends	No	No	Yes	Yes
$Overspend_{i,t}$	No	No	No	Yes
Observations	1192	1192	1192	1192
Adjusted R^2	0.896	0.898	0.947	0.947

All estimates in the table are of the order of magnitude 10^{-2} . The table reports the same set of regressions as table 1 but with the dependent variable as the average agricultural wages during the non-agricultural/lean season (Jan-Jun).

5.4.4 Robustness to using coefficient of variation to measure volatility

Since the standard deviation of expenditure may not necessarily be comparable across districts, we normalize $\sigma_{i,t}$ with the average program expenditure for a district, μ_i , which yields an alternative measure of expenditure volatility: the coefficient of variation $CV_{i,t}$. Analogous to table 1, table 11 reports the results obtained by replacing $\sigma_{i,t}$ and $e_{i,t} \times \sigma_{i,t}$ with $CV_{i,t}$ and $e_{i,t} \times CV_{i,t}$, respectively. In general, the results are consistent with the estimates reported in table 1. The coefficient of $e_{i,t}$ suggests wages to have increased by approximately 5.9% of the average wage at average expenditure and $CV_{i,t} = 0$. The interaction term suggests a wage compression of 32.5% at $\overline{CV}_{i,t}$ relative to $CV_{i,t} = 0$. While the coefficient of the interaction term suggests a more significant wage compression, the relatively higher standard errors comfortably envelop the wage compression estimate of 12.6% computed through estimates reported in table 1.

Table 11: Measuring volatility using the coefficient of variation

	(1)	(2)	(3)	(4)
$e_{i,t}$	1.761 [1.233]	2.467 [1.322]	1.960 [1.189]	2.197 [1.193]
$CV_{i,t}$	5.638 [4.781]	6.067 [5.060]	1.969 [4.832]	2.273 [4.825]
$e_{i,t} \times CV_{i,t}$	-0.015 [0.011]	-0.023 [0.014]	-0.022 [0.013]	-0.024 [0.013]
Fixed effects	Yes	Yes	Yes	Yes
Other Controls	No	Yes	Yes	Yes
District time trends	No	No	Yes	Yes
$Overspend_{i,t}$	No	No	No	Yes
Observations	1206	1206	1206	1206
Adjusted R^2	0.895	0.896	0.946	0.946

All estimates in the table are of the order of magnitude 10^{-2} . The table reports the same set of regressions as in table 1 but after replacing $\sigma_{i,t}$ and $e_{i,t} \times \sigma_{i,t}$ with $CV_{i,t}$ and $e_{i,t} \times CV_{i,t}$, respectively. $CV_{i,t} = 100 * \sigma_{i,t} / \mu_i$

5.4.5 Robustness to interacting the level and variability of expenditure with a time trend

In order to allow the effects of expenditure and its standard deviation to vary with time, we include $e_{i,t} * year$ and $\sigma_{i,t} * year$ as controls. In table 12 below we show the results. Column 1 is the same as our preferred specification in table 1. In columns 2, 3 and 4, we add $e_{i,t} * year$, $\sigma_{i,t} * year$ and both of them, respectively. We find that the coefficient of the interaction between $e_{i,t}$ and $\sigma_{i,t}$ is robust to the addition of these regressors.

Table 12: Controlling for time trend interacted with $e_{i,t}$ and $\sigma_{i,t}$

	(1)	(2)	(3)	(4)
$e_{i,t}$	2.406 [0.751]	4.460 [1.970]	2.086 [0.683]	6.065 [3.810]
$\sigma_{i,t}$	-3.142 [2.664]	-3.849 [2.778]	1.041 [3.989]	-7.800 [8.637]
$e_{i,t} \times \sigma_{i,t}$	-0.006 [0.002]	-0.005 [0.002]	-0.005 [0.002]	-0.005 [0.002]
$e_{i,t} \times t$		-0.277 [0.211]		-0.457 [0.436]
$\sigma_{i,t} \times t$			-0.510 [0.452]	0.426 [0.904]
All controls	Yes	Yes	Yes	Yes
Observations	1206	1206	1206	1206
Adjusted R^2	0.946	0.946	0.946	0.946

All estimates are of the order 10^{-2} . The table reports the full specification from table one with additional regressors.

6 Conclusion

In this article, we develop a theoretical model to show that variability in the level of program provision and the ensuing uncertainty compresses wage increases that occur due to public workfare programs. Consistent with our theory, we present empirical evidence to show that greater variability in program provision results in a larger compression of wage increases that resulted from public employment programs in India. Further, we find that compression of wage increases is more severe in districts where inflation is

low relative to where inflation is high.

The study has important policy implications. First, we show that incomplete and variable program implementation can diminish the increase in wages which results in an unambiguous welfare loss. This is especially a concern for developing economies which usually lack proper institutions or political will to implement large-scale programs. Second, our findings suggest that simply reducing the variability in program provision, without necessarily increasing the average expenditure, can be welfare enhancing. These findings are hence very relevant for program design as well as pre-implementation cost-benefit analysis.

While not a part of this study, a promising area for future research is to understand how employers and workers expectations about public workfare programs affect labor market outcomes. This is important if the continued existence of the workfare alleviates uncertainty regarding the level of program provision in the future. Such a revision in expectations will increase wages simply due to the accompanying decompression of wage increases. This may provide an alternative explanation to the delayed effect of public employment programs on wages that has been noted in the recent empirical literature on employment guarantees in India.³¹

³¹See for e.g. [Berg et al. \(2018\)](#), [Bahal \(2016\)](#), and [Shrivastava \(2015\)](#) who find wages to increase due to the overall exposure to the workfare as opposed to a contemporaneous jump in wages.

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Appendix

Proof of Proposition 1

The employer's problem is

$$\max_{w_t, w_{t+1}} V(r_t, w_t, r_{t+1}, w_{t+1}) = \pi_1(r_t, w_t) + \pi_1(r_{t+1}, w_{t+1})$$

$$s.t. \quad w_t \leq w_{t+1}$$

The Lagrangian can be written as

$$\mathcal{L} = \pi_1(r_t, w_t) + \pi_1(r_{t+1}, w_{t+1}) - \lambda(w_t - w_{t+1})$$

F.O.C.

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial w_t} &= \pi_1^{w_t}(r_t, w_t) - \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial w_{t+1}} &= \pi_1^{w_{t+1}}(r_{t+1}, w_{t+1}) + \lambda = 0 \end{aligned}$$

When the constraint doesn't bind, i.e. $w_{t+1} \geq w_t$ then $\lambda = 0$, and the optimal wages are $w_t = w_1^*(r_t)$, $w_{t+1} = w_1^*(r_{t+1})$. Since $\frac{dw_1^*(r)}{dr} < 0$, therefore we can see that the constraint will *not* bind with these optimal wages only when $r_{t+1} \leq r_t$

Therefore, when $r_{t+1} > r_t$, the constraint will bind. Let $r_{t+1} = kr_t$, where $k > 1$. When the constraint binds, $\lambda > 0$ and $w_t = w_{t+1}$. Hence the objective function becomes.

$$V(r_t, r_{t+1}, w_t) = \pi_1(r_t, w_t) + \pi_1(r_{t+1}, w_t)$$

Finding the derivative w.r.t. w_t

$$\begin{aligned} V^{w_t}(r_t, r_{t+1}, w_t) &= \pi_1^{w_t}(r_t, w_t) + \frac{\partial}{\partial w_t} [f(r_{t+1}l_0(w_t)) - w_t r_{t+1} l_0(w_t)] \\ &= \pi_1^{w_t}(r_t, w_t) + \frac{\partial}{\partial w_t} [f(kr_t l_0(w_t)) - w_t kr_t l_0(w_t)] \\ &= \pi_1^{w_t}(r_t, w_t) + kr_t (f'(kr_t l_0(w_t)) l_0'(w_t) - l_0(w_t) - w_t l_0'(w_t)) \\ &< \pi_1^{w_t}(r_t, w_t) + kr_t (f'(r_t l_0(w_t)) l_0'(w_t) - l_0(w_t) - w_t l_0'(w_t)) \\ &(\because k > 1 \text{ \& } f''(\cdot) < 0) \\ &= \pi_1^{w_t}(r_t, w_t) + k\pi_1^{w_t}(r_t, w_t) \end{aligned}$$

At $w_t = w_1^*(r_t)$, the right hand side is zero. Hence,

$$w_t^*(r_t, r_{t+1}) = \arg \max_{w_t} V(r_t, r_{t+1}, w_t) < w_1^*(r_t) \quad \text{for } r_{t+1} > r_t$$

Proof of Proposition 2

Let the rationing rate in the first period be fixed at $r_t = \bar{r}$. Let the rationing rate in the second period (r_{t+1}) take one of two possible values - $r_a > \bar{r}$ with probability p and $r_b \leq \bar{r}$ with probability $1 - p$.

When $r_{t+1} = r_a$, the wage constraint binds and the total profit over two periods is

$$V_a(\bar{r}, r_a, w_t) = \pi_1(\bar{r}, w_t) + \pi_1(r_a, w_t)$$

When $r_{t+1} = r_b$, the constraint does not bind and total profit is

$$V_b(\bar{r}, r_b, w_t, w_{t+1}) = \pi_1(\bar{r}, w_t) + \pi_1(r_b, w_{t+1})$$

Now the expected total profit is

$$V(\bar{r}, r_a, r_b, w_t, w_{t+1}, p) = pV_a + (1 - p)V_b$$

Differentiating this with respect to w_t , we can see from the proof of proposition 1 that when the first period wage is at the statically optimum value, i.e. $w_t = w_1^*(\bar{r})$, then $V_a^{w_t}(\bar{r}, r_a, w_t) < 0$. We can also see that since the second term of V_b is independent of w_t . therefore $V_b^{w_t}(\bar{r}, r_b, w_t, w_{t+1}) = 0$. Therefore, $V^{w_t}(\bar{r}, r_a, r_b, w_t, w_{t+1}, p) < 0$ for any $p > 0$. Therefore, the dynamically optimal first period wage

$$w_t^*(\bar{r}, r_b, p) = \arg \max_{w_t} V(\bar{r}, r_a, r_b, w_t, w_{t+1}, p) < w_1^*(r_t) \quad \text{for } p > 0$$

Proof of Proposition 3

The sum of current profit and future expected profit given by

$$V(m, x, r_t, w_t) = \pi(r_t, w_t) + \frac{1}{x} \left(\int_{m-x/2}^{\bar{r}_{t+1}(w_t)} \pi(r_{t+1}, w_1^*(r_{t+1})) dr_{t+1} + \int_{\bar{r}_{t+1}(w_t)}^{m+x/2} \pi(r_{t+1}, w_t) dr_{t+1} \right)$$

Differentiating this with w_t , we get

$$\begin{aligned} V^{w_t}(m, x, r_t, w_t) = & \pi^{w_t}(r_t, w_t) + \frac{1}{x} \left(\pi(\bar{r}_{t+1}(w_t), w_1^*(\bar{r}_{t+1}(w_t))) \frac{\partial \bar{r}_{t+1}(w_t)}{\partial w_t} \right. \\ & \left. - \pi(\bar{r}_{t+1}(w_t), w_t) \frac{\partial \bar{r}_{t+1}(w_t)}{\partial w_t} + \int_{\bar{r}_{t+1}(w_t)}^{m+x/2} \pi^{w_t}(r_{t+1}, w_t) dr_{t+1} \right) \end{aligned}$$

We can see that the second and third term will cancel out and we are left with

$$V^{w_t}(m, x, r_t, w_t) = \pi^{w_t}(r_t, w_t) + \frac{1}{x} \int_{\bar{r}_{t+1}(w_t)}^{m+x/2} \pi^{w_t}(r_{t+1}, w_t) dr_{t+1} \quad (A1)$$

It is evident that the second integral will be negative as the wage is higher than the optimum wage at that rationing level. i.e. for $r_{t+1} > \bar{r}_{t+1}(w_t)$, $w_t > w_1^*(r_{t+1})$ and therefore $\pi^{w_t}(r_{t+1}, w_t) < 0$.

At the dynamically optimal wage $w_t^*(m, x, r_t)$, this derivative will be zero.

$$V^{w_t}(m, x, r_t, w_t^*(m, x, r_t)) = 0$$

Now we use the implicit function theorem with respect to x

$$\frac{\partial w_t^*(m, x, r_t)}{\partial x} = - \frac{V^{w_t x}}{V^{w_t w_t}} \Big|_{w_t=w^*}$$

Hence, $\frac{\partial w_t^*(m, x, r_t)}{\partial x} < 0$ iff $V^{w_t x}|_{w_t=w^*} < 0$ ($\because V^{w_t w_t}|_{w_t=w^*} < 0$).

$$V^{w_t x} = 0 - \frac{1}{x^2} \int_{\bar{r}_{t+1}(w_t)}^{m+x/2} \pi^{w_t}(r_{t+1}, w_t) dr_{t+1} + \frac{1}{x} \pi^{w_t}(m+x/2, w_t) \frac{d(m+x/2)}{dx}$$

Therefore, $V^{w_t x} < 0$ implies

$$\begin{aligned} & -\frac{1}{x^2} \int_{\bar{r}_{t+1}(w_t)}^{m+x/2} \pi^{w_t}(r_{t+1}, w_t) dr_{t+1} + \frac{1}{2x} \pi^{w_t}(m+x/2, w_t) < 0 \\ \iff & \frac{x}{2} \pi^{w_t}(m+x/2, w_t) < \int_{\bar{r}_{t+1}(w_t)}^{m+x/2} \pi^{w_t}(r_{t+1}, w_t) dr_{t+1} \end{aligned}$$

However, using equation **A1** we know that at the dynamically optimal wage:

$$\int_{\bar{r}_{t+1}(w_t)}^{m+x/2} \pi^{w_t}(r_{t+1}, w_t) dr_{t+1} = -x\pi^{w_t}(r_t, w_t)$$

Therefore, at $w_t = w_t^*$

$$\frac{x}{2}\pi^{w_t}(m+x/2, w_t) < -x\pi^{w_t}(r_t, w_t)$$

Since $x > 0$, this implies

$$\pi^{w_t}(r_t, w_t) < -\frac{\pi^{w_t}(m+x/2, w_t)}{2}$$

Summary statistics

Table A1: Summary statistics

	Mean	SD	25%ile	75%ile	Obs.
$w_{i,t}$	56.93	27.25	40.87	61.19	1340
$e_{i,t}$	152.09	176.64	47.52	180.76	1340
$e_{i,t}^a$	195.33	232.45	57.02	232.78	1340
$\sigma_{i,t}$	50.38	79.50	7.82	52.46	1206
$\sigma_{i,t}^a$	63.47	102.00	6.63	75.40	1206
$\pi_{i,t}$	6.82	4.29	2.98	9.94	1206

Note: $w_{i,t}$: Average daily field wage (2001 prices). $e_{i,t}$: per capita program expenditure (2001 prices). $e_{i,t}^a$: per capita fund allocation at the start of a financial year (2001 prices). $\sigma_{i,t}$: Standard deviation of expenditure. $\sigma_{i,t}^a$: Standard deviation of fund allocation. $\pi_{i,t}$: Year-on-year growth rate of CPI-RL.

Suggestive evidence towards market power

As described in section 3.2, even if labor markets are perfectly competitive, compression of wage increases can happen through a ratcheting mechanism. Wages increase in response to the upward pressure created by a workfare but downward rigidity stops them from adjusting downwards when program provision worsens. This implies that the average effect of workfare spending would be lower if the maximum level of workfare spending in the past has been high. We construct the following specification to test if this mechanism is at play.

$$w_{i,t} = \beta_1 e_{i,t} + \beta_2 e_{i,t}^{max} + \beta_3 (e_{i,t} \times e_{i,t}^{max}) + \alpha_i + \gamma_t + \xi_{it} + \beta \mathbf{X}_{i,t} + Overspend_{i,t} + \varepsilon_{i,t} \quad (A2)$$

Here, $e_{i,t}^{max}$ is the maximum expenditure in district i in years $\{2001, \dots, t-1\}$. This is calculated recursively for each year starting from 2002, analogous to the way $\sigma_{i,t}$ is calculated. If the ratcheting mechanism is at work, then β_1 and β_2 should be positive, while β_3 should be negative.

The results are shown in column 2 of table A2. Column 1 is the result of estimating equation 3, presented here for comparison. In column 2, both β_1 and β_3 have the expected signs but are statistically insignificantly different from 0. This suggests that the ratcheting mechanism is likely not responsible for compression of wage increases. Since this is the only mechanism through which wage compression can occur when

markets are perfectly competitive, this result adds to suggestive evidence discussed in [subsection 4.1](#) against the presence of perfect competition in agricultural labor markets in India. Indirectly, this suggests wage compression to be a result of the forward-looking behavior of employers' with wage setting power discussed in [section 2.2](#).

Table A2: Testing for Market Power vs Perfect Competition

	(1)	(2)
$e_{i,t}$	2.406 [0.751]	0.487 [0.441]
$\sigma_{i,t}$	-3.142 [2.664]	
$e_{i,t} \times \sigma_{i,t}$	-0.006 [0.002]	
$e_{i,t}^{max}$		-0.007 [1.034]
$e_{i,t} \times e_{i,t}^{max}$		-0.002 [0.001]
All controls	Yes	Yes
Observations	1206	1206
Adjusted R^2	0.946	0.945

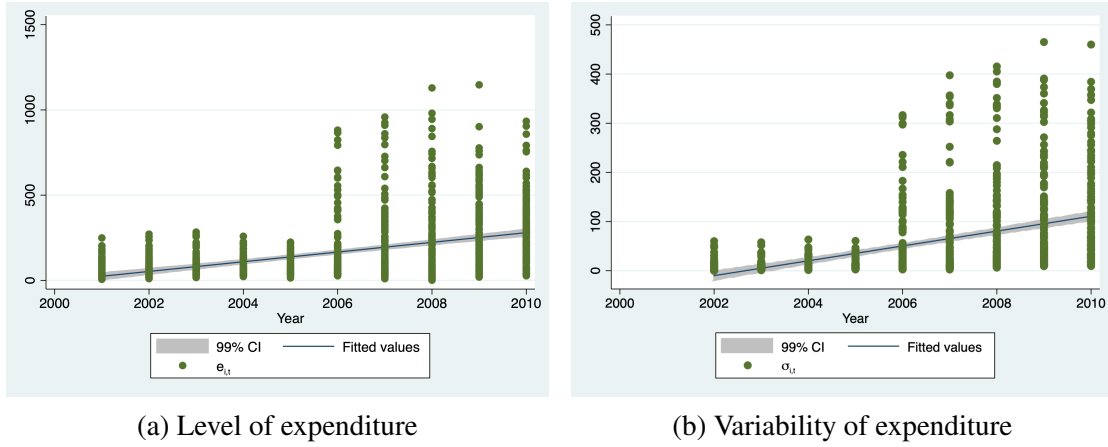
All estimates in the table are of the order of magnitude 10^{-2} . Column 1 is the preferred specification reported in column 4 of [table 1](#). Column 2 reports results obtained by estimating [equation A2](#) with the same set of additional controls as in column 1. Here, $e_{i,t}^{max}$ is the maximum expenditure in district i in years $\{2001, \dots, t-1\}$. This is calculated recursively for each year starting from 2002, analogous to the way $\sigma_{i,t}$ is calculated. Standard errors are clustered at the district level and are reported in parentheses.

The above result is consistent with an important finding noted in multiple studies (see, for e.g., [Berg et al., 2018](#); [Shrivastava, 2015](#); [Bahal, 2016](#)). Private wages do not increase immediately after large increases in workfare expenditure following the introduction of NREGA. If the labor supply channel in a perfectly competitive market was at play, this should not be the case. The time-lag involved in wage increases noted in the literature is also indicative of a forward-looking behavior where wage-setting employers withhold substantive and immediate wage increases to avoid the downward wage rigidity constraint in the event program provision deteriorates in the future.

Time trend in level and variability of expenditure

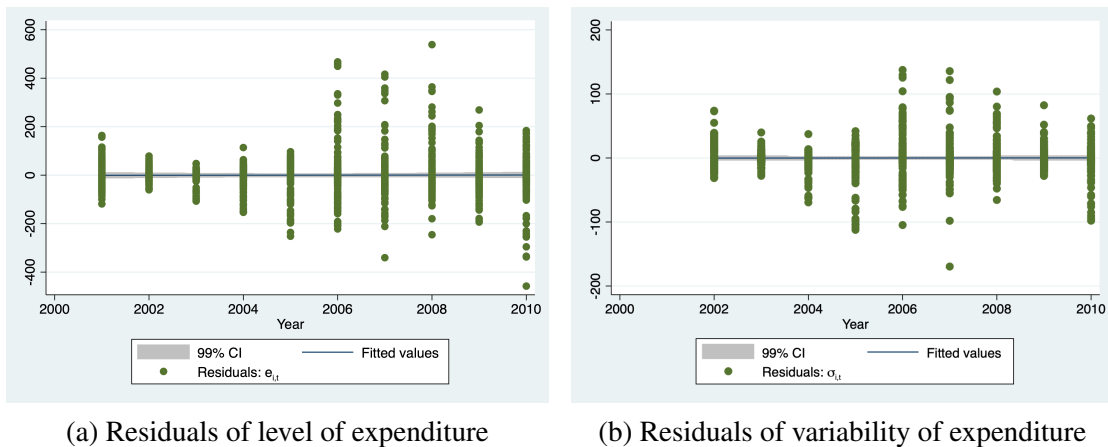
Since NREGA was implemented in phases and it was a larger program than SGRY, the level and variance of expenditure do change mechanically over time (see figure A1). It is precisely for this reason we control for year fixed effects and district-specific time trends. In figure A2 we show scatter plots of the residuals of the level and variability of expenditure, after controlling for fixed effects and time trends, and as expected, they are not correlated with time.

Figure A1: Trend of $e_{i,t}$ and $\sigma_{i,t}$ over time



Note: The graph on the left shows the time trend in the variable $e_{i,t}$. The one on the right shows the time trend in the variable $\sigma_{i,t}$.

Figure A2: Trend of $e_{i,t}$ and $\sigma_{i,t}$ over time after partialling out fixed effects and district time trends



Note: The graph on the left shows the time trend in the residuals after regressing $e_{i,t}$ on district and year fixed effects and district-specific time trends). The one on the right shows the time trend in the residuals after regressing $\sigma_{i,t}$ on district and year fixed effects and district-specific time trends.

Regressions with nominal wage

Here we conduct the same set of regressions in tables 1 and 2 but with nominal wages instead of real wages as the outcome variable. The results are presented in tables A3 and A4, which are analogous to tables 1 and 2 but with nominal wages as the regressand. We find that the results are very similar to those with real wages, i.e., we find that increased variability in spending compresses increases in nominal wage and that the compression is larger in districts with lower inflation.

Table A3: Effect of Variability in Expenditure on Nominal Wages

	(1)	(2)	(3)	(4)
$e_{i,t}$	4.446 [1.490]	5.171 [1.670]	3.530 [1.310]	4.104 [1.329]
$\sigma_{i,t}$	-3.504 [3.700]	-6.883 [3.606]	-5.116 [4.124]	-5.539 [4.207]
$e_{i,t} \times \sigma_{i,t}$	-0.008 [0.004]	-0.009 [0.005]	-0.007 [0.004]	-0.008 [0.004]
Fixed effects	Yes	Yes	Yes	Yes
Other Controls	No	Yes	Yes	Yes
District time trends	No	No	Yes	Yes
$Overspend_{i,t}$	No	No	No	Yes
Observations	1206	1206	1206	1206
Adjusted R^2	0.869	0.873	0.947	0.947

All estimates are of the order 10^{-2} . The table reports the same set of regressions as table 1 but with the dependent variable as the nominal agricultural wage.

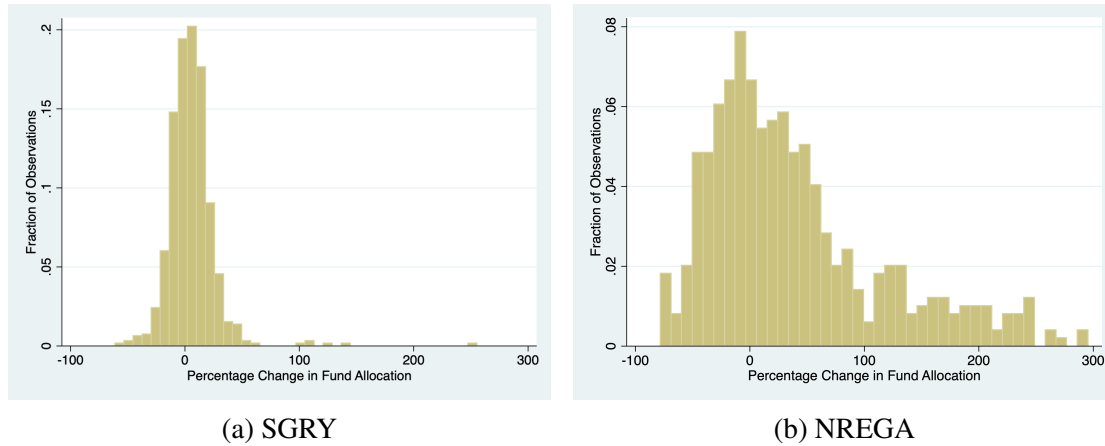
Table A4: Nominal wage compression under low and high inflation

	(1)	(2)	(3)	(4)	(5)
$e_{i,t}$	3.397 [1.821]	4.140 [2.026]	3.568 [1.598]	4.230 [1.607]	3.295 [1.371]
$\sigma_{i,t}$	-1.685 [3.665]	-5.077 [3.671]	-4.796 [4.465]	-5.313 [4.558]	
$e_{i,t} \times \sigma_{i,t}$	-0.006 [0.005]	-0.008 [0.005]	-0.007 [0.005]	-0.008 [0.004]	-0.010 [0.004]
$e_{i,t} \times \sigma_{i,t} \times \pi_i$ (Low)	-0.088 [0.054]	-0.080 [0.052]	-0.028 [0.039]	-0.029 [0.039]	-0.010 [0.006]
$e_{i,t} \times \pi_i$ (Low)	12.267 [7.209]	11.566 [7.019]	2.682 [4.435]	2.641 [4.394]	
$\sigma_{i,t} \times \pi_i$ (Low)	7.056 [10.628]	5.258 [10.376]	1.519 [10.744]	1.777 [10.772]	
Fixed effects	Yes	Yes	Yes	Yes	Yes
Other Controls	No	Yes	Yes	Yes	Yes
District time trends	No	No	Yes	Yes	Yes
$Overspend_{i,t}$	No	No	No	Yes	Yes
Observations	1206	1206	1206	1206	1206
Adjusted R^2	0.872	0.875	0.947	0.947	0.947

All estimates in the table are of the order of magnitude 10^{-2} . The table reports the same set of regressions as table 2 but with the dependent variable as the nominal agricultural wage.

Distribution of changes in fund allocation

Figure A3: Distribution of Fund Allocation Changes



Note: **Figure A3(a)** shows the distribution of year-on-year percentage change in real per capita fund allocation made under SGRY. Data is for 442 districts from 2001-2007 (2230 observations). **Figure A3(b)** shows the distribution of year-on-year percentage change in real per capita fund allocation made under NREGA. Data is for 442 districts from 2006-2010 (1615 observations). For both figures, the unit of observation is a district-year and fund allocation changes are top coded at 300%.